BTC options vs Kalshi

Generated 2026-10-03 23:13 UTC. Are prediction markets efficient once you account for the risk premium in options? Option-implied probabilities (Deribit) vs Kalshi's 17:00 New York daily Bitcoin markets. Research only; the paper-trading log never sends orders.

Findings

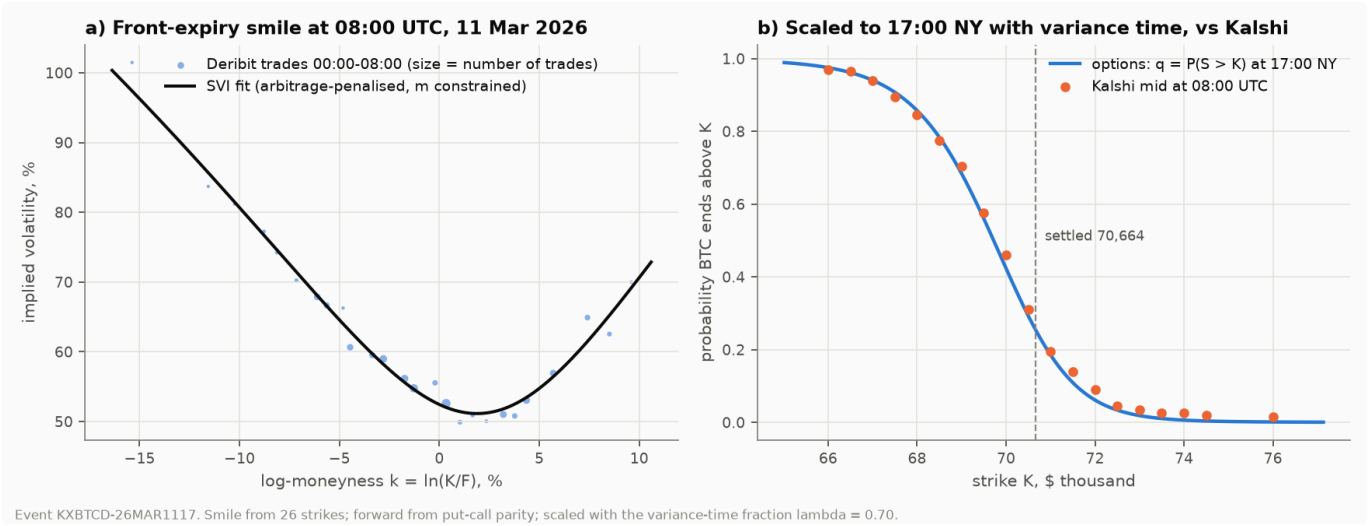
0.99 ± 0.04
logit slope of option q (calibrated = 1)

1.14 ± 0.04
logit slope of the Kalshi mid, 08:00 UTC

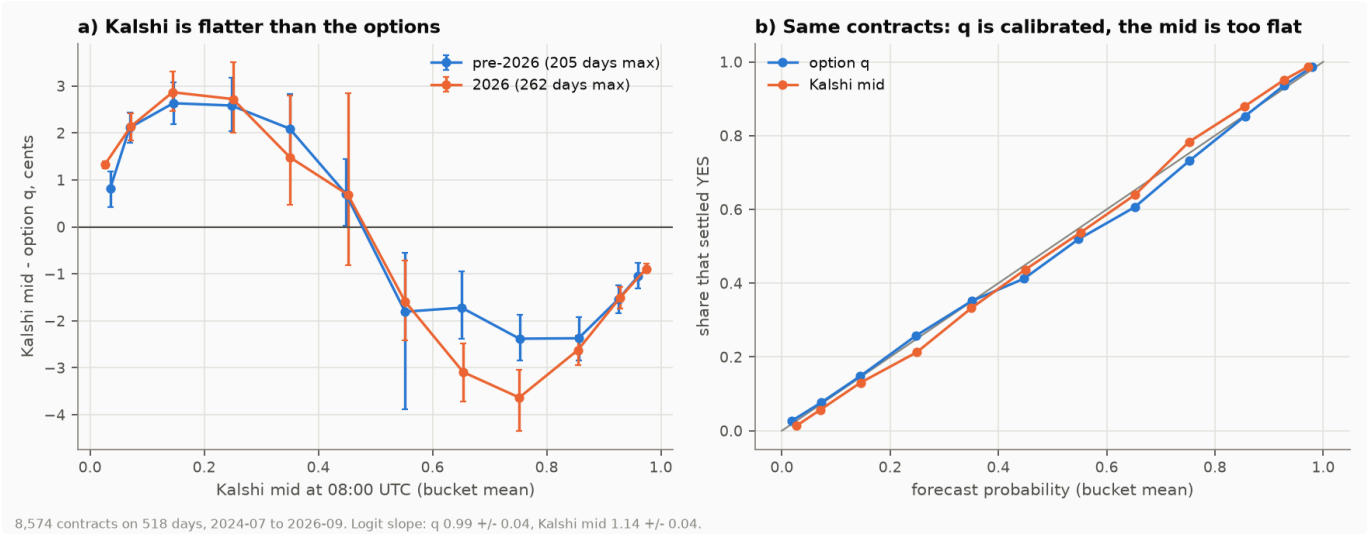
0.0962 / 0.0962
Brier: Kalshi mid / option q (518 days)

≈ 0¢
maker edge per signal with simulated fills, 2026

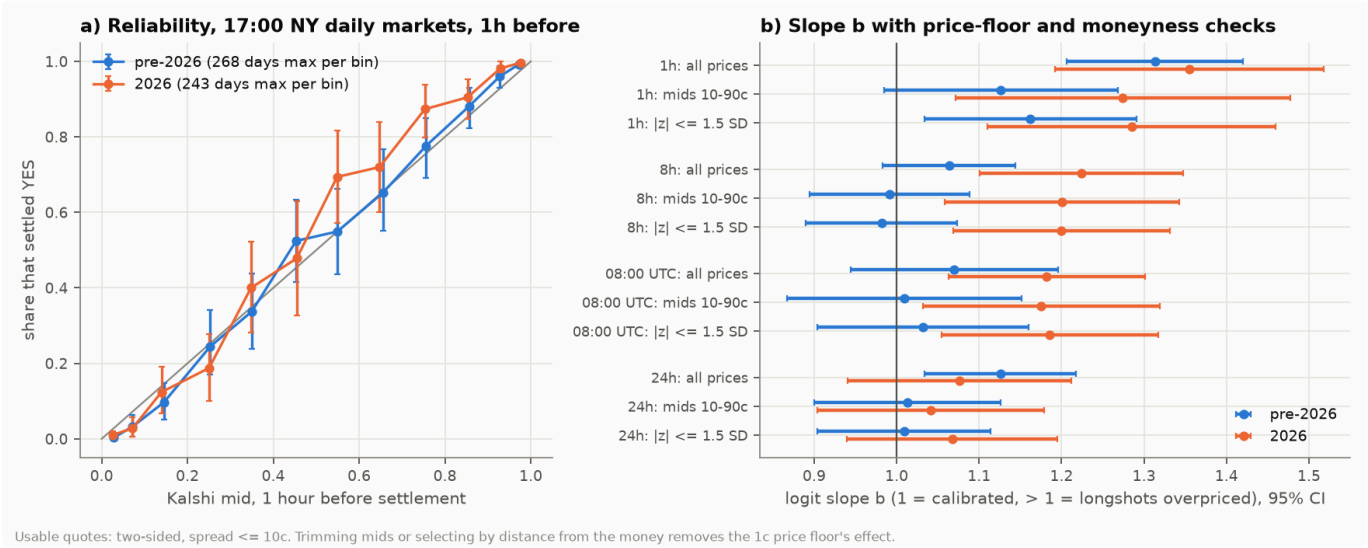
1. **The options get it right.** The 1-day option probability q is calibrated on Kalshi's strikes.
2. **Kalshi has a favourite-longshot bias in 2026** (slope 1.18–1.36). It survives the 1¢ price-floor check; before 2026 it was mostly that floor.
3. **Kalshi is flatter than the options by 2–4¢, but just as informative**: equal Brier scores, similar information with fresh option data.
4. **The gap can't be traded.** Takers lose to fees and the spread; maker profits vanish once fills are simulated.
5. **Kalshi is efficient up to its costs.** The bias coincides with more takers buying longshots in 2026 (suggestive, not significant).



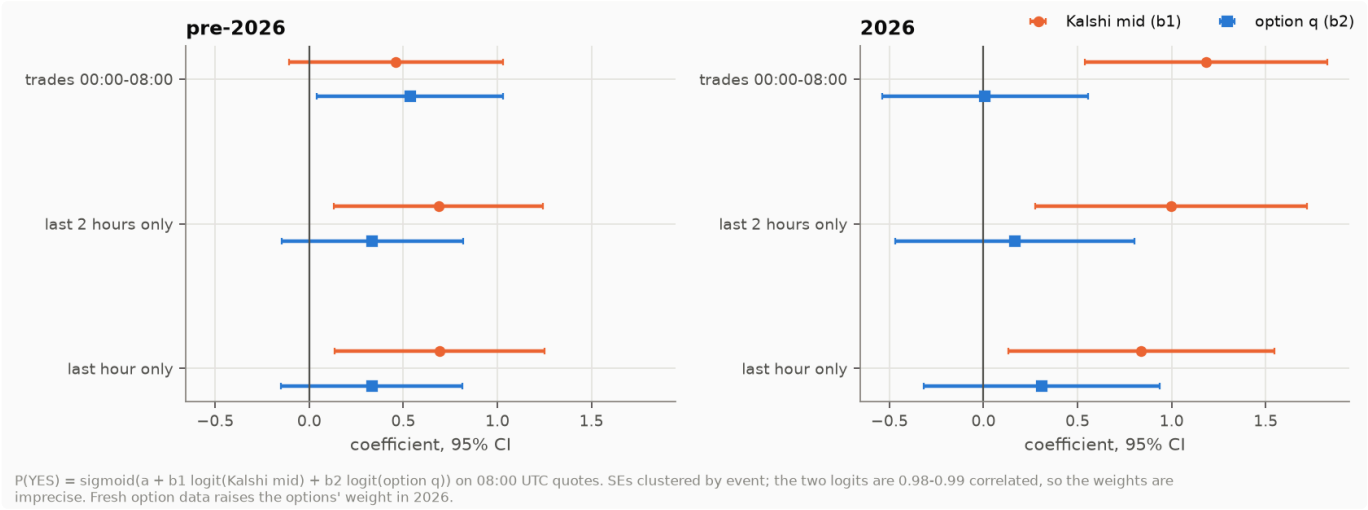
From option trades to a 1-day probability. One day: Deribit trades and the fitted SVI smile at 08:00 UTC, then q = P(S > K) at 17:00 NY next to Kalshi's mids. (figures/report/fig1_option_to_probability.png)



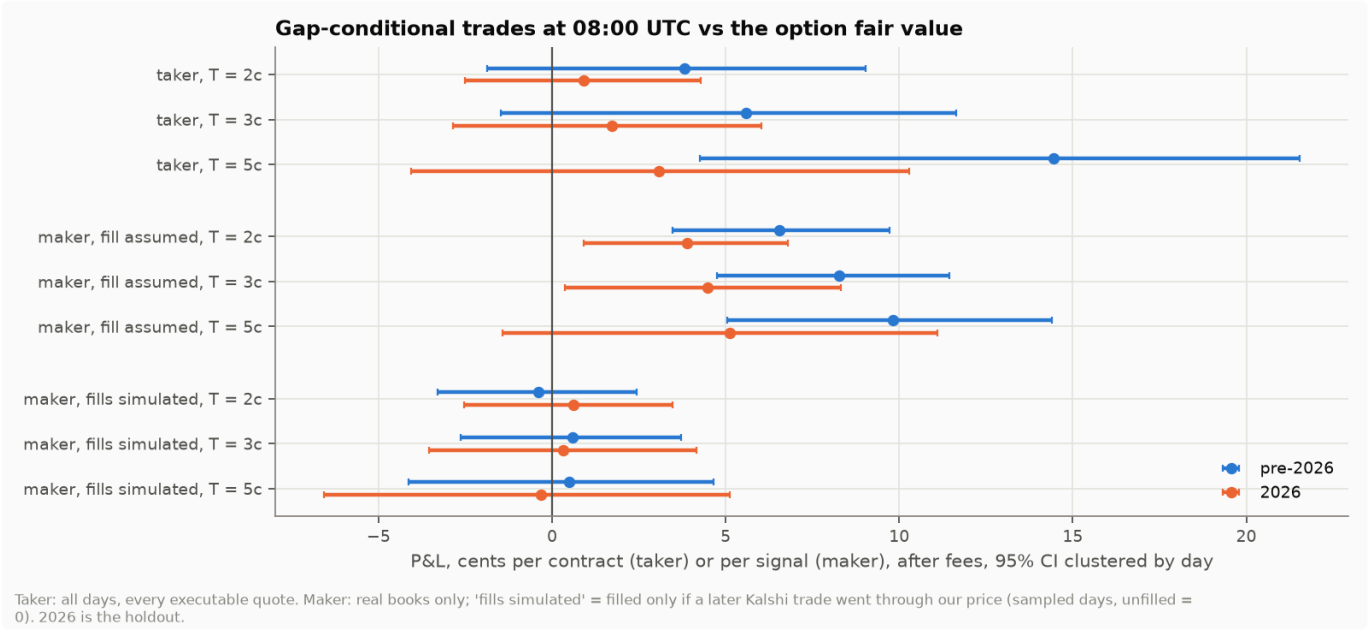
Kalshi is flatter than the options. Kalshi mid minus option q by price bucket, and reliability of both on the same 8,574 contracts: q is calibrated, the mid too flat. (figures/report/fig4_kalshi_vs_options.png)



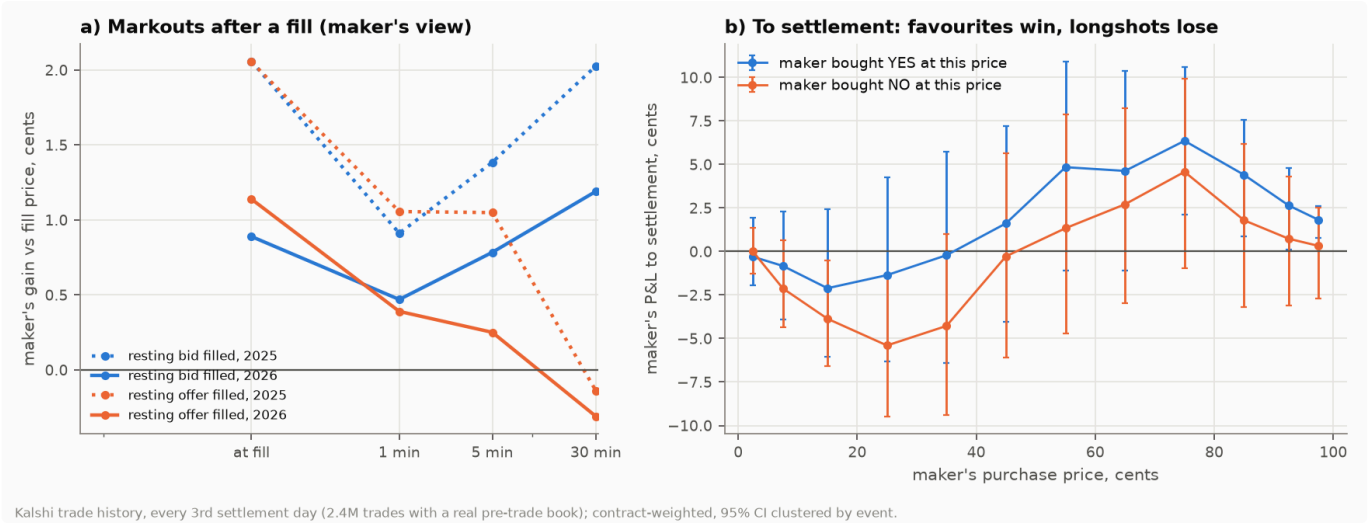
Kalshi calibration and the price-floor check. Reliability 1h before settlement, and the favourite-longshot slope b with mids trimmed or selected by distance from the money. (figures/report/fig3_kalshi_calibration.png)



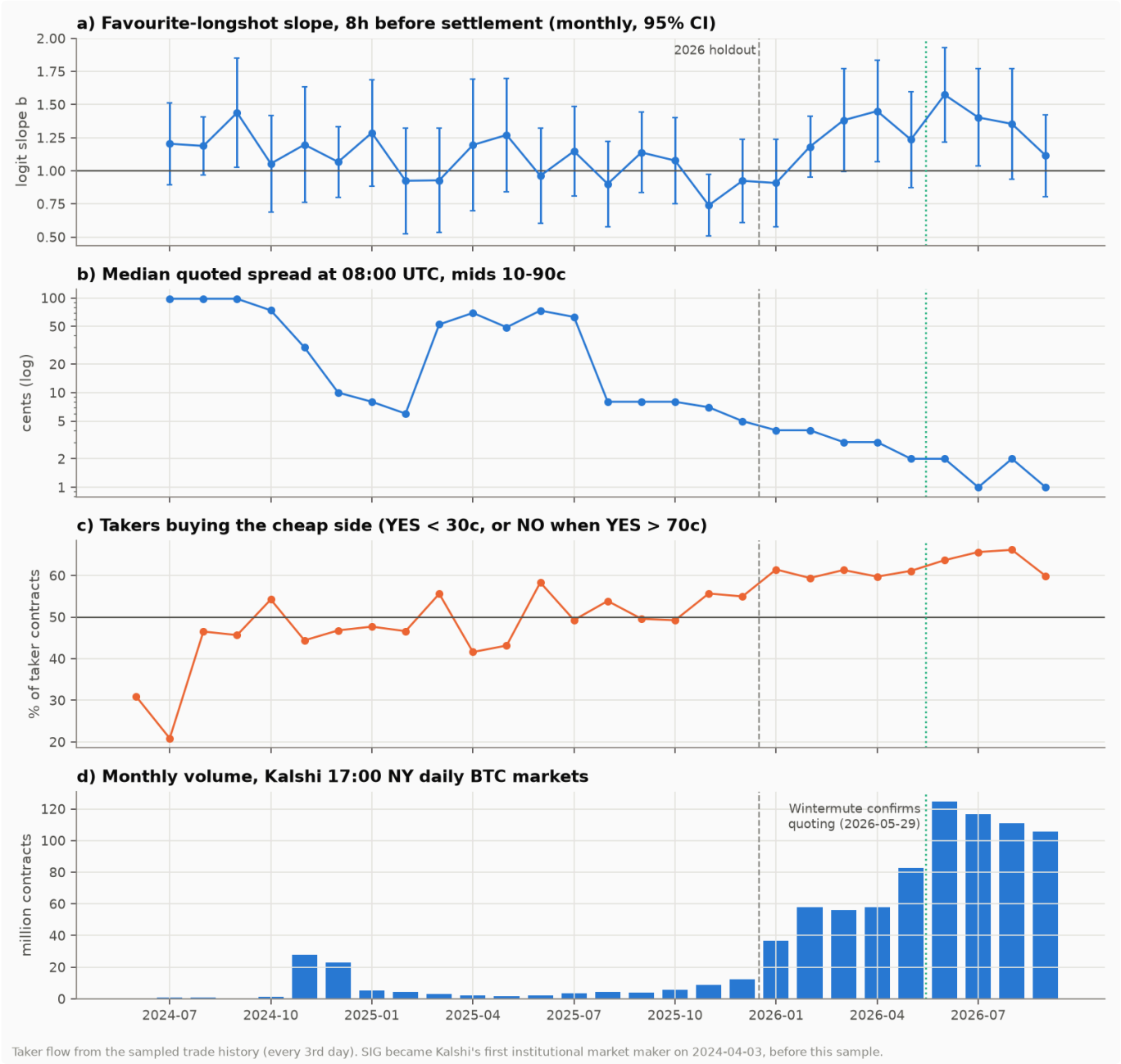
Which price carries the information? Encompassing weights of the Kalshi mid and option q, with the usual 8-hour smile and with smiles from only the last 2 hours or 1 hour. (figures/report/fig5_encompassing_staleness.png)



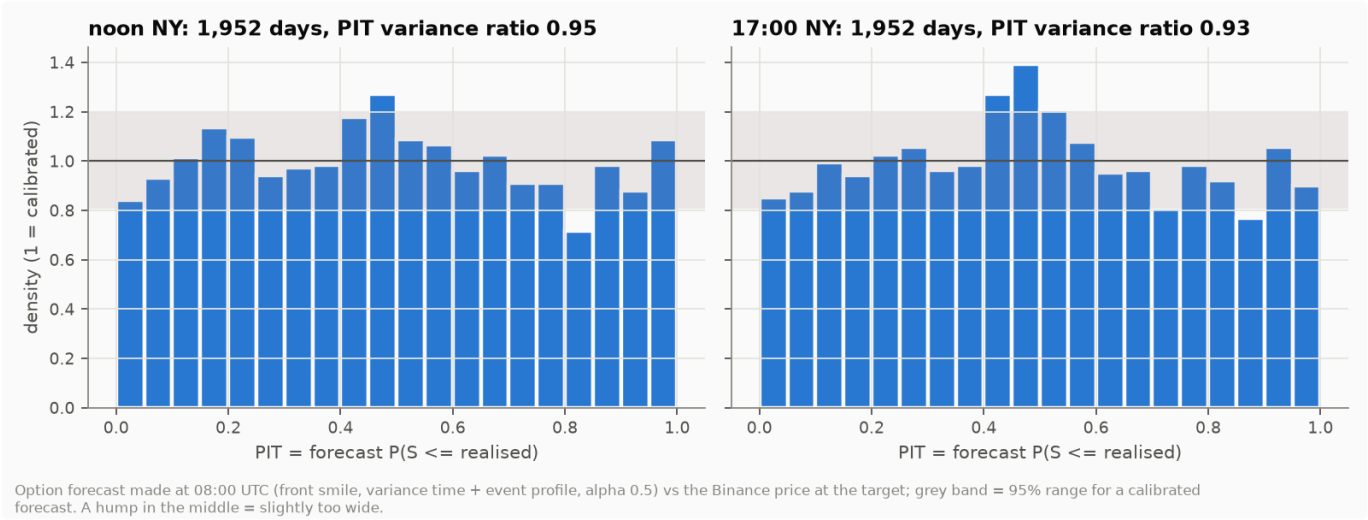
Can the gap be traded? Taker and maker P&L with 95% CIs. The maker edge disappears once fills are simulated from Kalshi's trade history. (figures/report/fig6_trade_tests.png)



What resting orders earn. Markouts after a fill (maker's view) and P&L to settlement by purchase price: favourites win, longshots lose. (figures/report/fig7_markouts.png)



Bias over time and market structure. Monthly slope b, quoted spread, taker longshot buying and volume, with the 2026 holdout and the Wintermute date. (figures/report/fig8_bias_over_time.png)



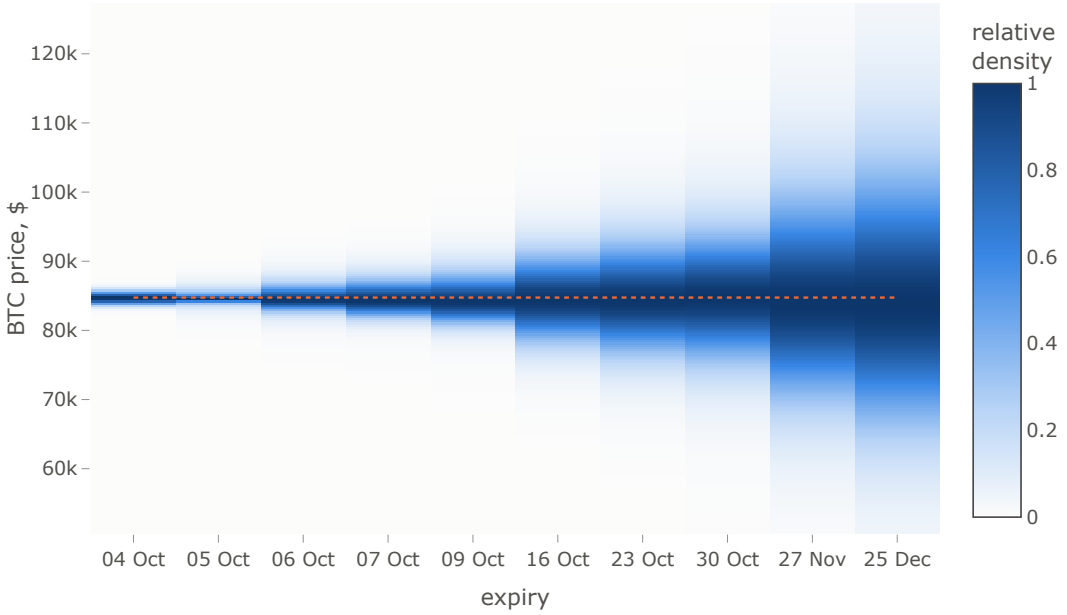
The 1-day option forecasts are calibrated. PIT histograms on 1,952 days: flat = calibrated; the small central hump = slightly too wide (a small variance premium). (figures/report/fig2_short_horizon_validation.png)

Live: options and Kalshi today, paper trading

Current option-implied distributions

Deribit snapshot 2026-10-03 23:10 UTC; index 84,748. SVI fitted to mark IVs per expiry, density scaled to its own maximum.

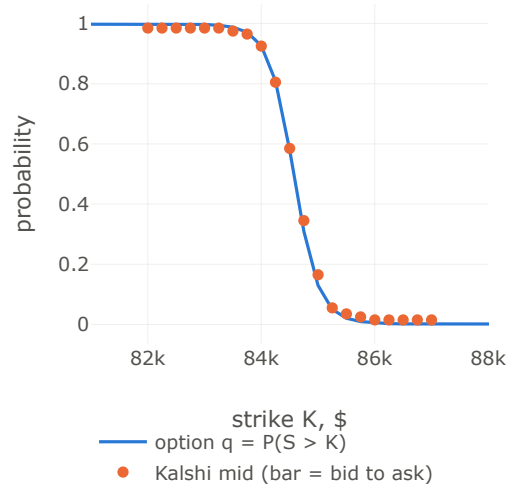
Option-implied density by expiry



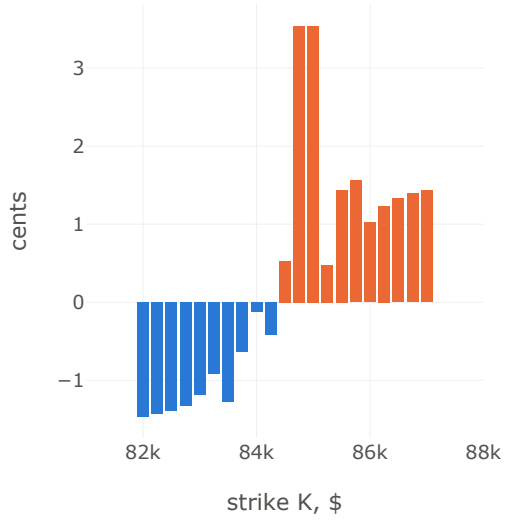
Today: Kalshi vs options at 08:00 UTC

Kalshi 08:00 UTC snapshot on 2026-10-03; settles 2026-10-03 21:00 UTC; option forward 84,573.

P(BTC above K at settlement)



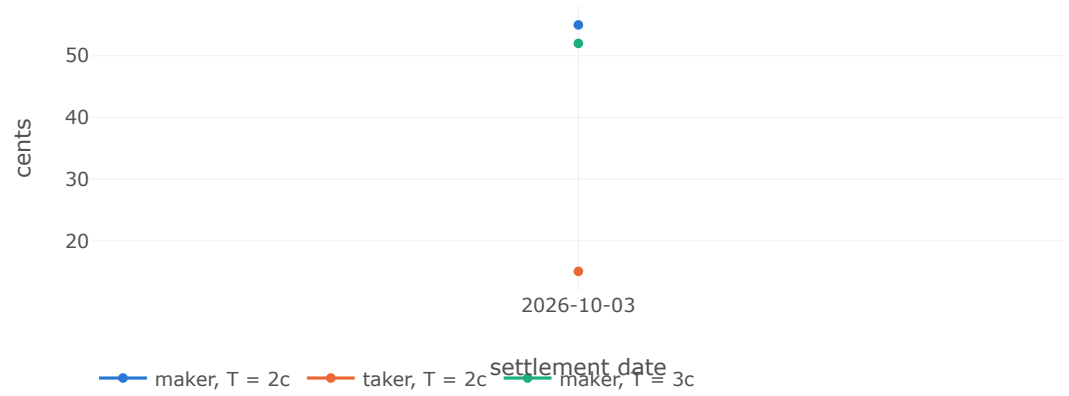
Kalshi mid - option q



Paper-trading log

Gap rule at 08:00 UTC for T = 2, 3, 5¢, taker and maker. Maker fills from Kalshi's public trade history ("through" = a later trade went beyond our price). P&L in cents per contract after fees.

Cumulative paper P&L, cents per contract summed over signals (fills "through")



Settled signals by rule

T_c	style	signals	fill_rate	pnl_through_c	pnl_if_filled_c
2.0	maker	3	1.0	18.333	18.333
2.0	taker	1	1.0	15.050	15.050
3.0	maker	2	1.0	26.000	26.000

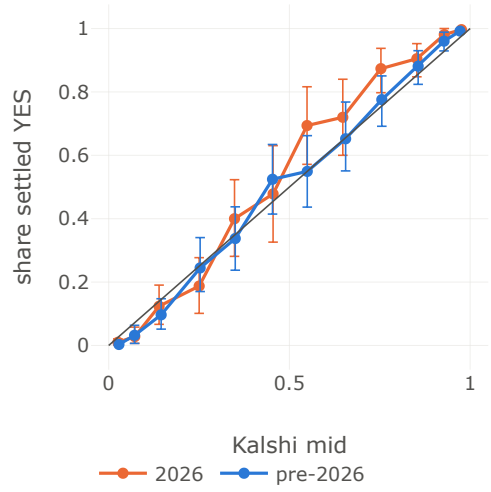
Latest signals

date	T_c	style	side	strike	q	yes_bid	yes_ask	price	edge_c	best_size	outcome	filled_through	pnl_c_through
2026-10-03	2.000	maker	NO	84,749.99	0.310	0.340	0.350	0.650	4.044	5,160.00	0.000	1.000	35.000
2026-10-03	2.000	maker	NO	84,999.99	0.130	0.160	0.170	0.830	4.044	5,677.00	0.000	1.000	17.000
2026-10-03	2.000	maker	NO	85,749.99	0.009	0.020	0.030	0.970	2.057	3,025.60	0.000	1.000	3.000
2026-10-03	2.000	taker	NO	84,999.99	0.130	0.160	0.170	0.840	2.094	450.00	0.000	1.000	15.050
2026-10-03	3.000	maker	NO	84,749.99	0.310	0.340	0.350	0.650	4.044	5,160.00	0.000	1.000	35.000
2026-10-03	3.000	maker	NO	84,999.99	0.130	0.160	0.170	0.830	4.044	5,677.00	0.000	1.000	17.000

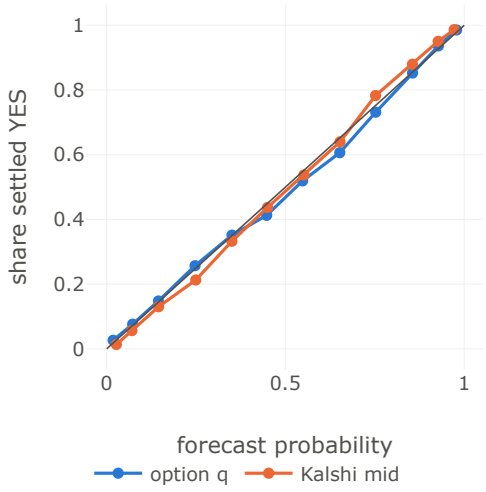
Kalshi analysis

Calibration (interactive)

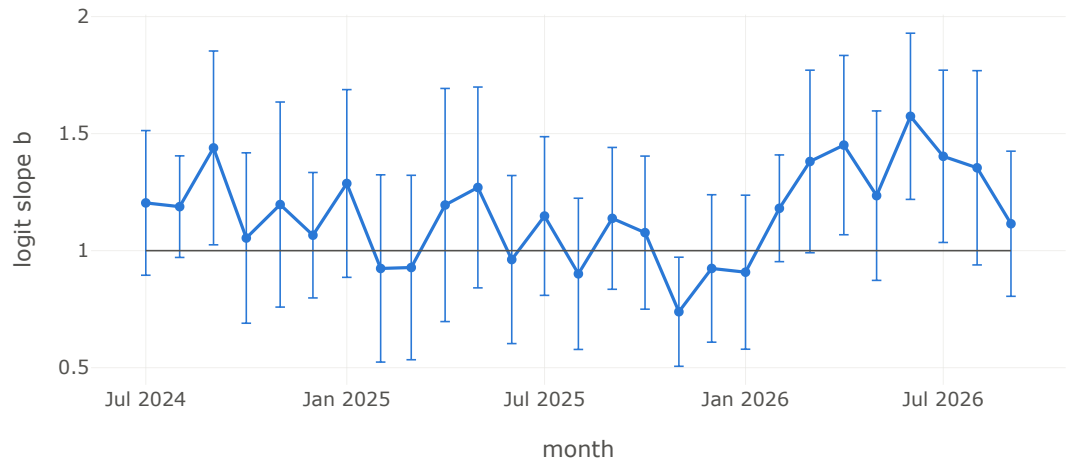
Kalshi reliability, 1h before (17:00 NY)

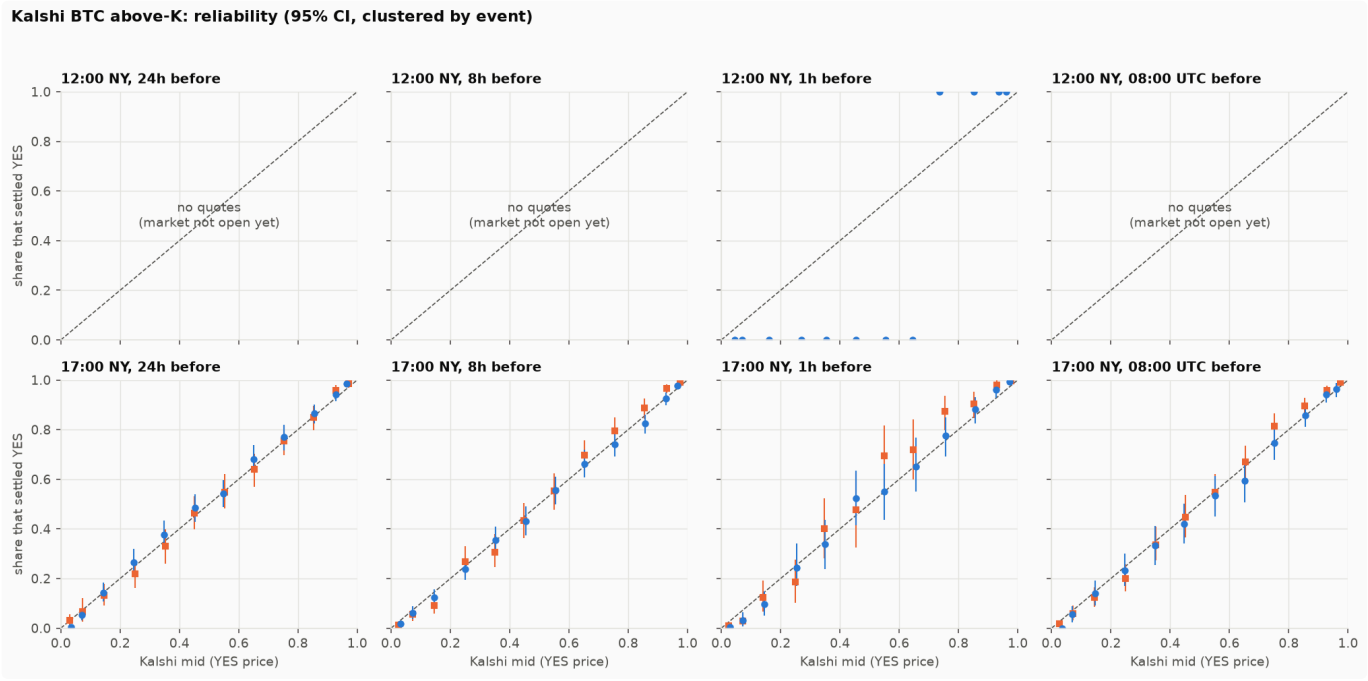


Same contracts at 08:00 UTC: q vs Kalshi

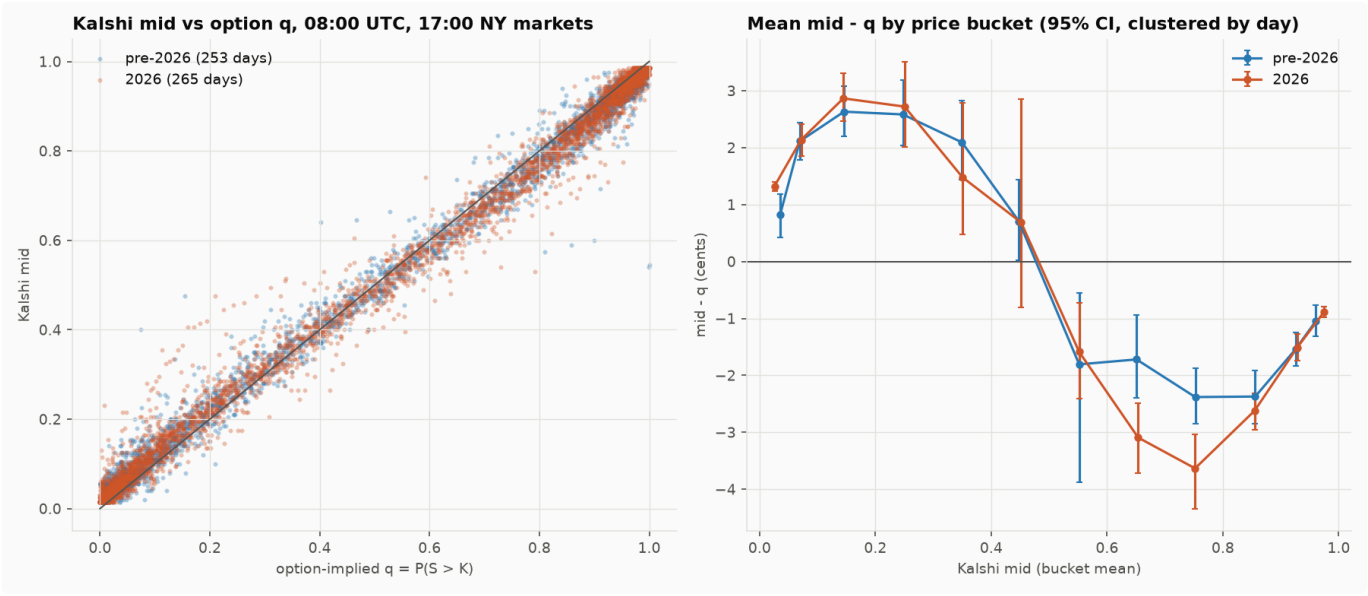


Monthly favourite-longshot slope (1 = calibrated, 95% CI)

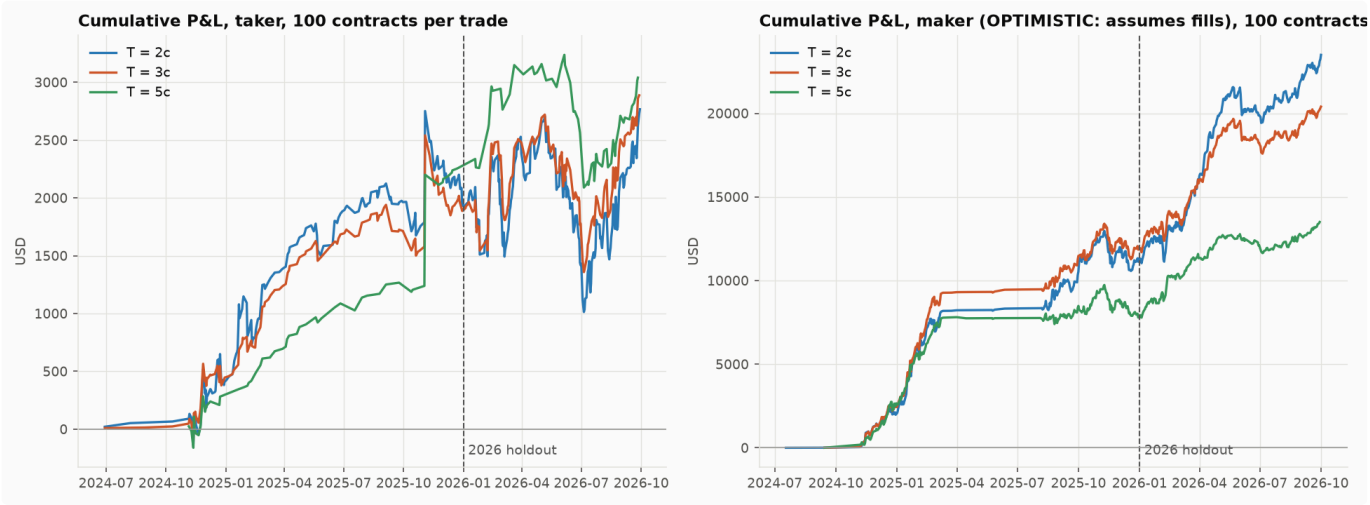




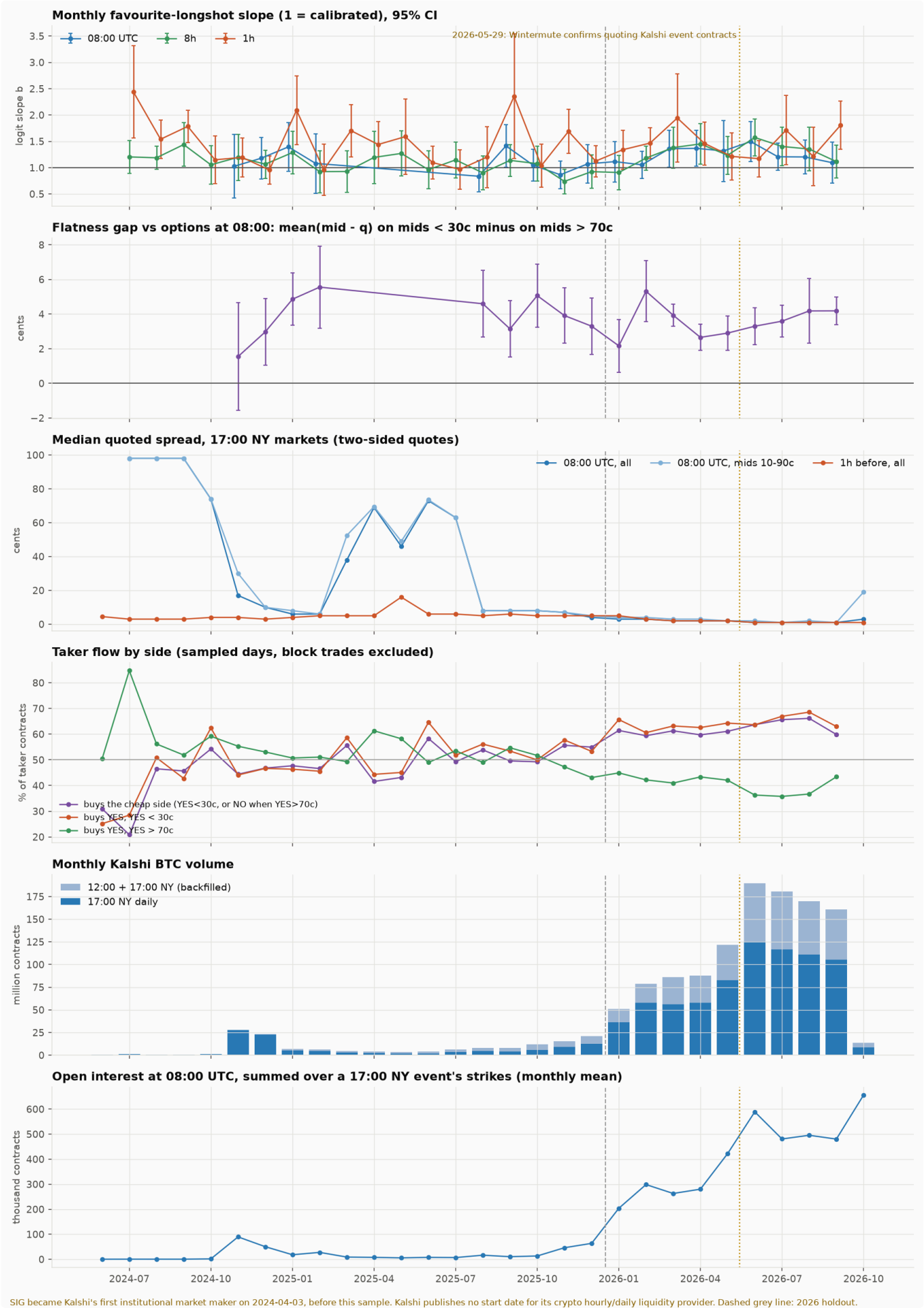
Reliability by quote time and period. Share settling YES against the Kalshi mid, 24h / 8h / 1h before and at 08:00 UTC, pre-2026 vs 2026. (pm/kalshi_calibration/reliability.png)



Kalshi mid vs option q, every contract. Scatter of mid against q at 08:00 UTC and the mean gap by bucket with day-clustered CIs. (pm/kalshi_vs_options/kalshi_vs_options.png)



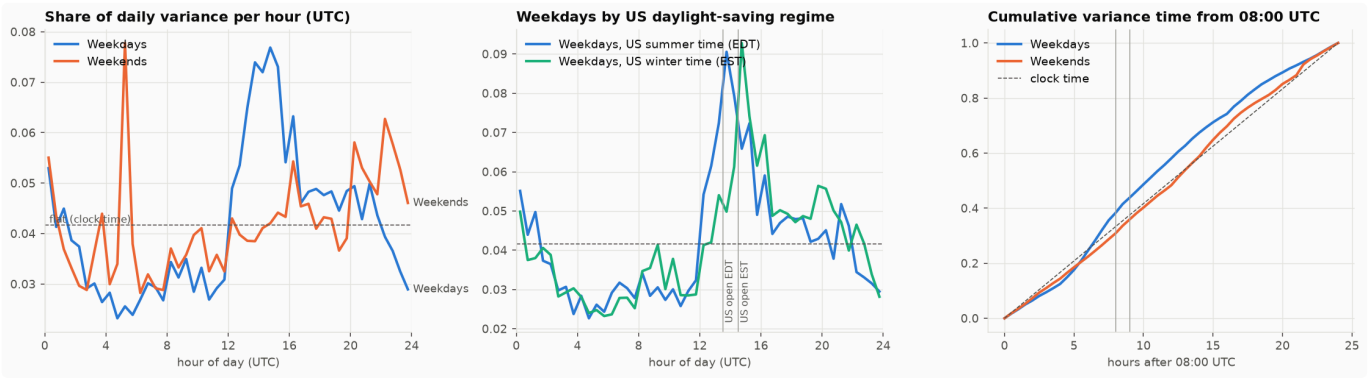
Gap rule: cumulative P&L. Taker and (optimistic, fill-assumed) maker P&L at T = 2, 3, 5c, 100 contracts per trade; dashed line = 2026 holdout. (pm/kalshi_gap_trade/cumulative_pnl.png)



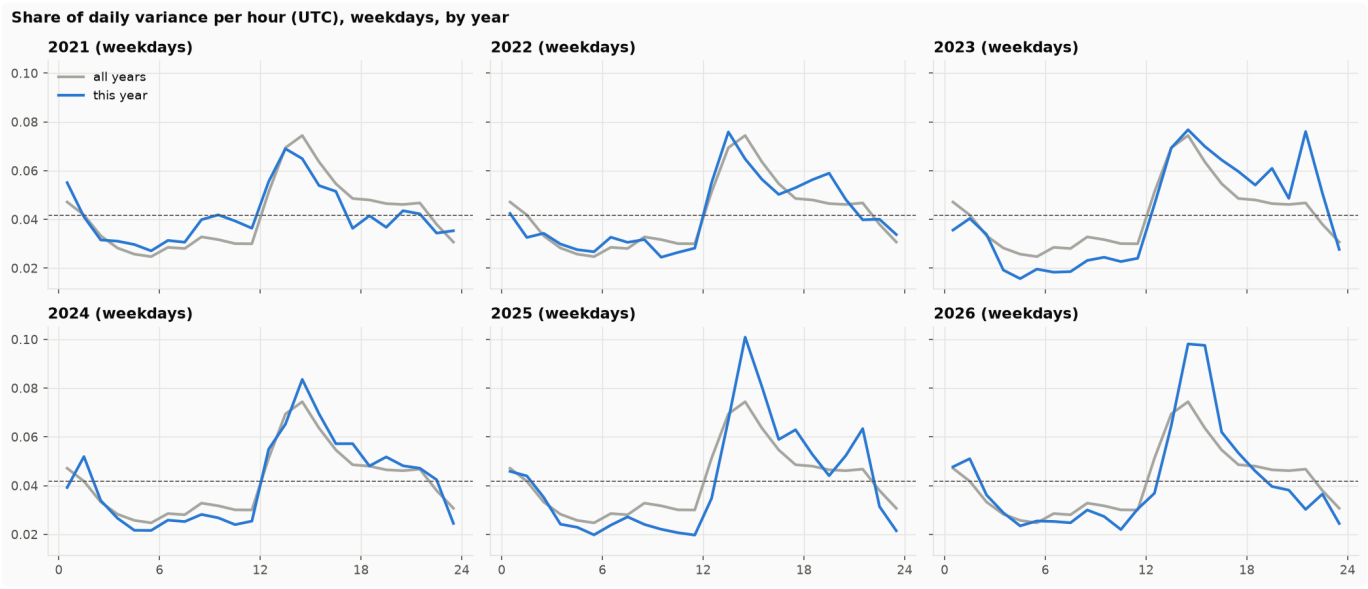
Bias vs activity, spreads and taker flow (full detail). Monthly slopes at three quote times, the gap vs options, spreads, taker flow by side, volume and open interest. (pm/kalshi_bias_time/bias_vs_activity.png)

1-day option forecasts

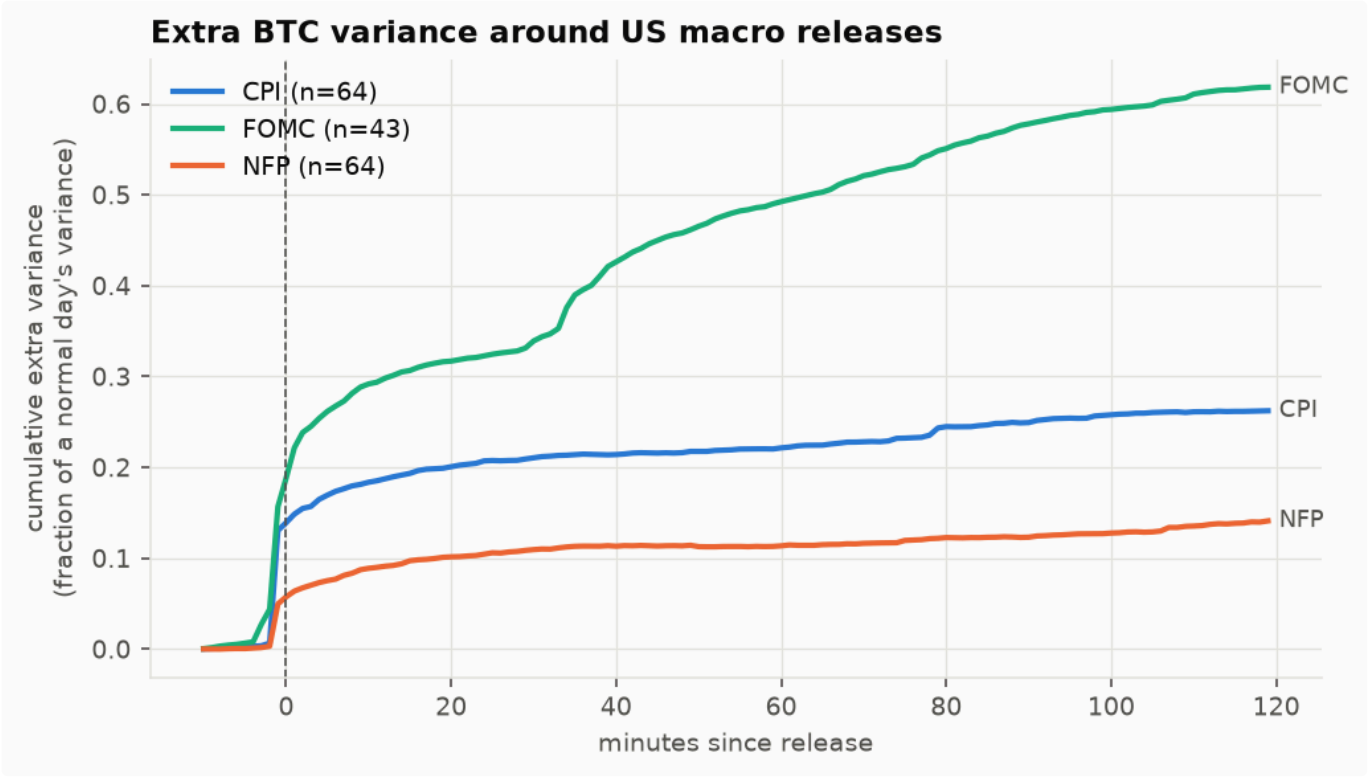
Scaling the 08:00 UTC front-expiry smile to noon and 17:00 New York with variance time and macro-event days; validated on 1,869 days.



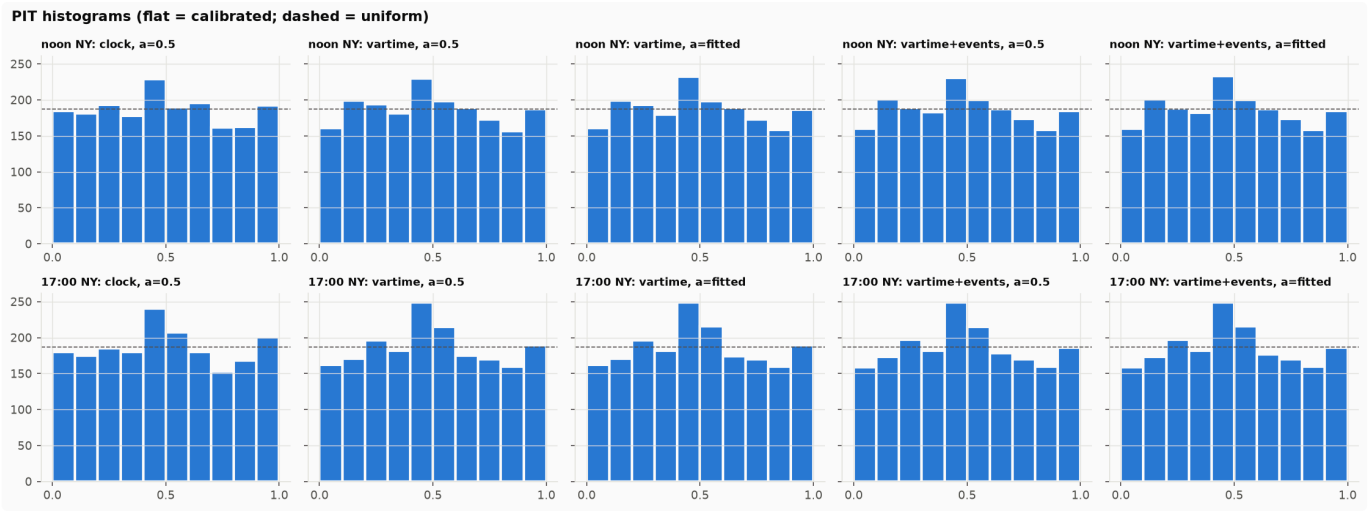
Intraday variance profile. Share of a day's variance per minute from squared 1-minute Binance returns, weekday vs weekend: the basis of variance time. (intraday/intraday_profile.png)



The profile by year. The intraday shape is stable across years; the weekend level falls from about 0.73 of a weekday in 2021. (intraday/intraday_by_year.png)



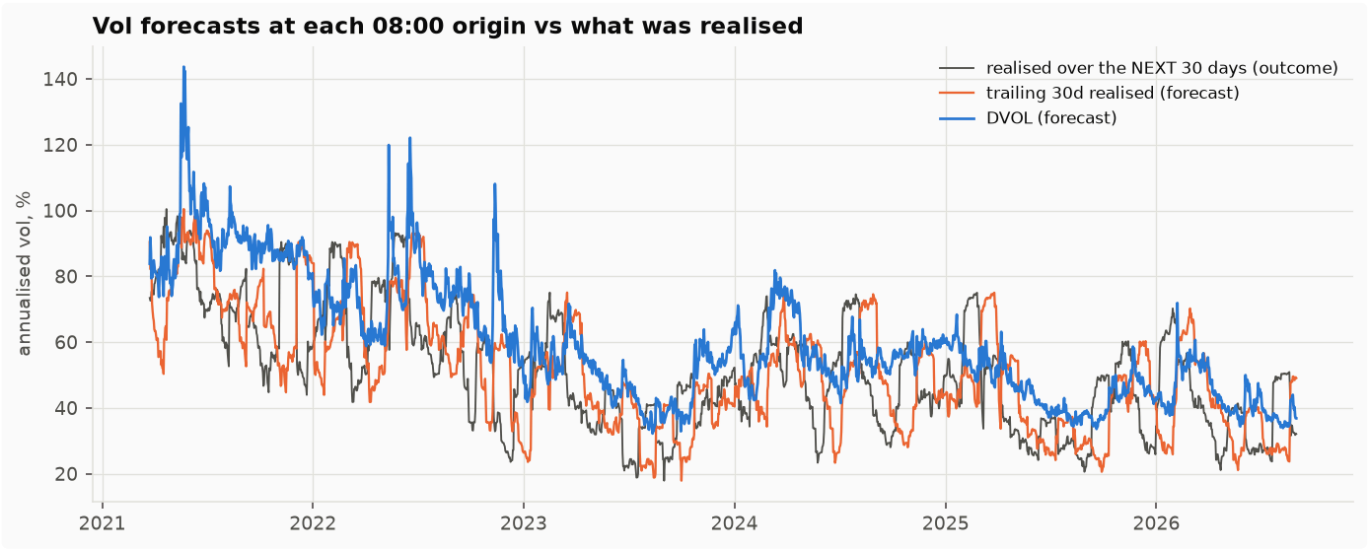
Macro-event days. Excess variance around CPI, NFP and FOMC releases, from earlier events only: most of it arrives within minutes. (short_horizon/event_excess.png)



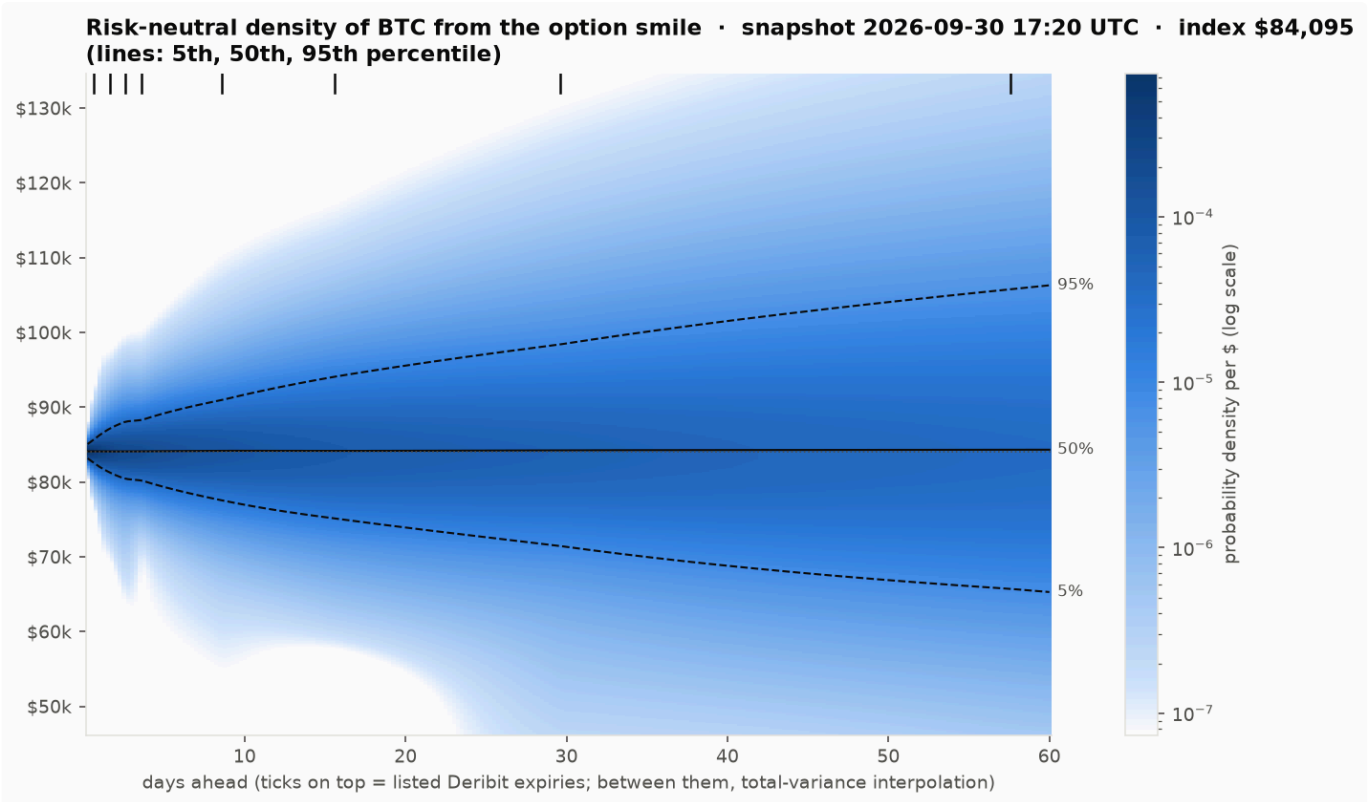
Validation of every variant. PIT histograms for clock vs variance time, with and without events, alpha 0.5 vs fitted. (short_horizon/pit_histograms.png)

30-day option forecasts (phases 2-3)

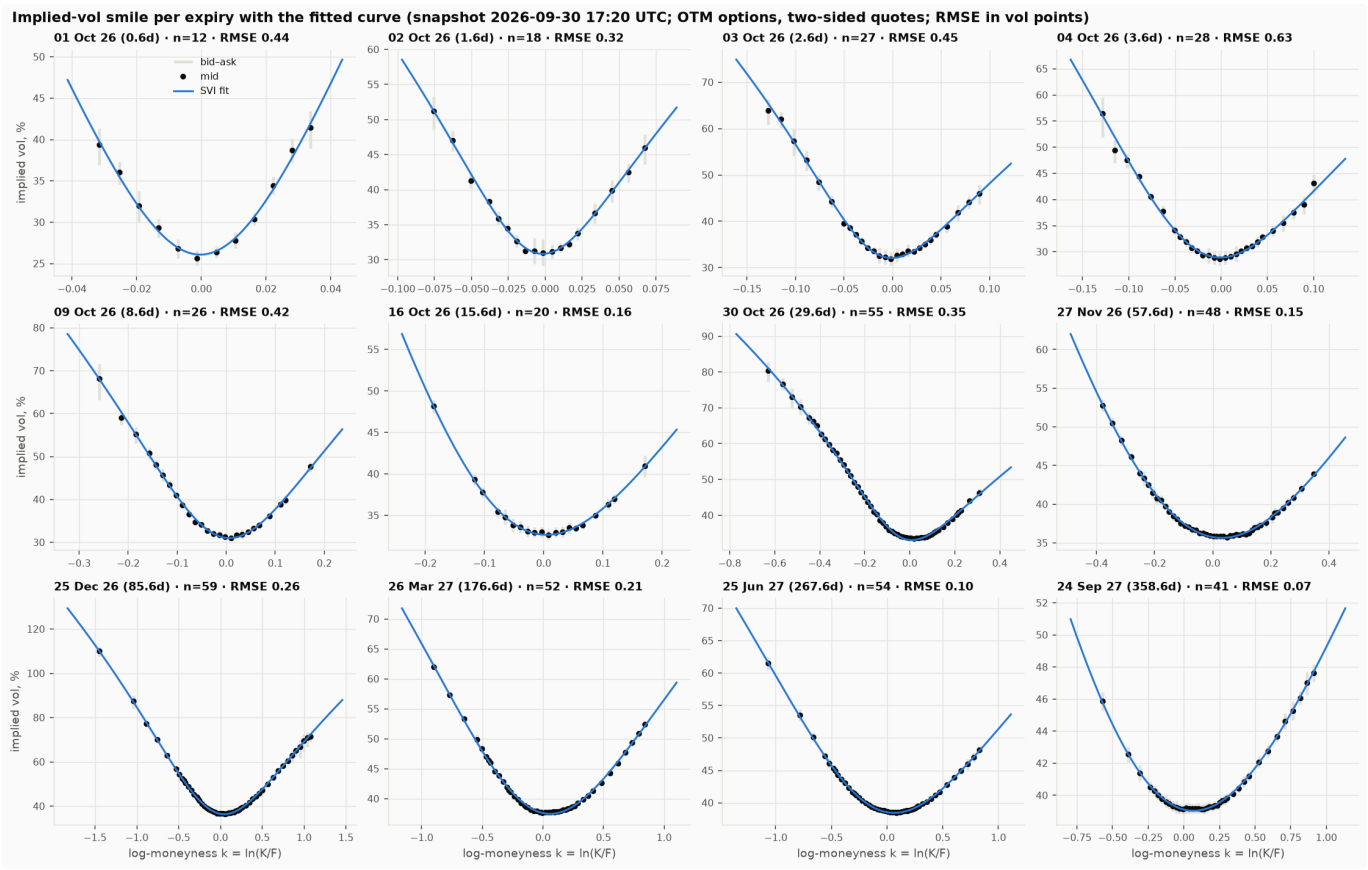
Phases 2–3: 30-day forecasts from DVOL and from the full option smile, scored against the Deribit delivery price.



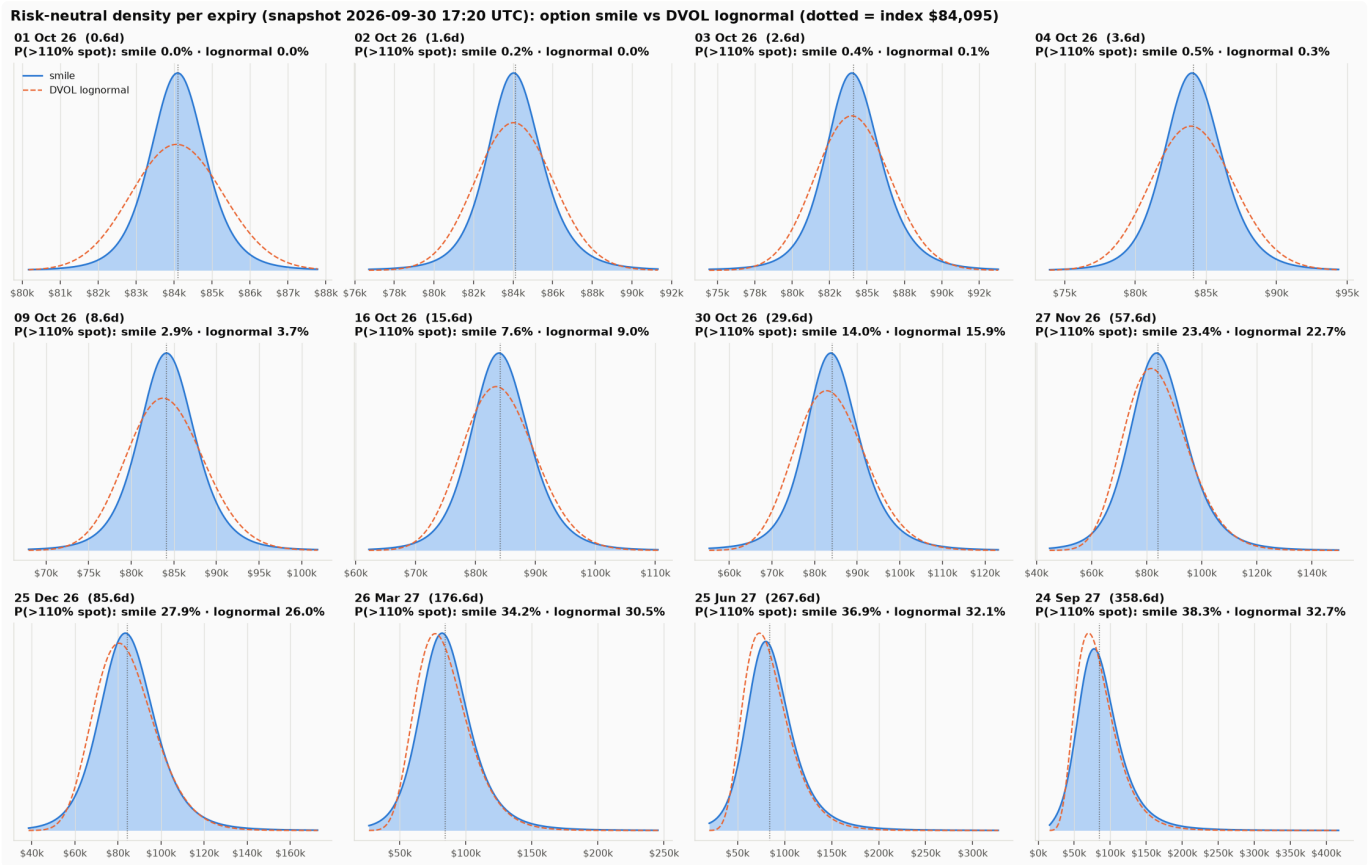
DVOL vs realised volatility. DVOL averaged about 60% against 51% realised over the next 30 days: a variance risk premium. (phase2/vol_levels.png)



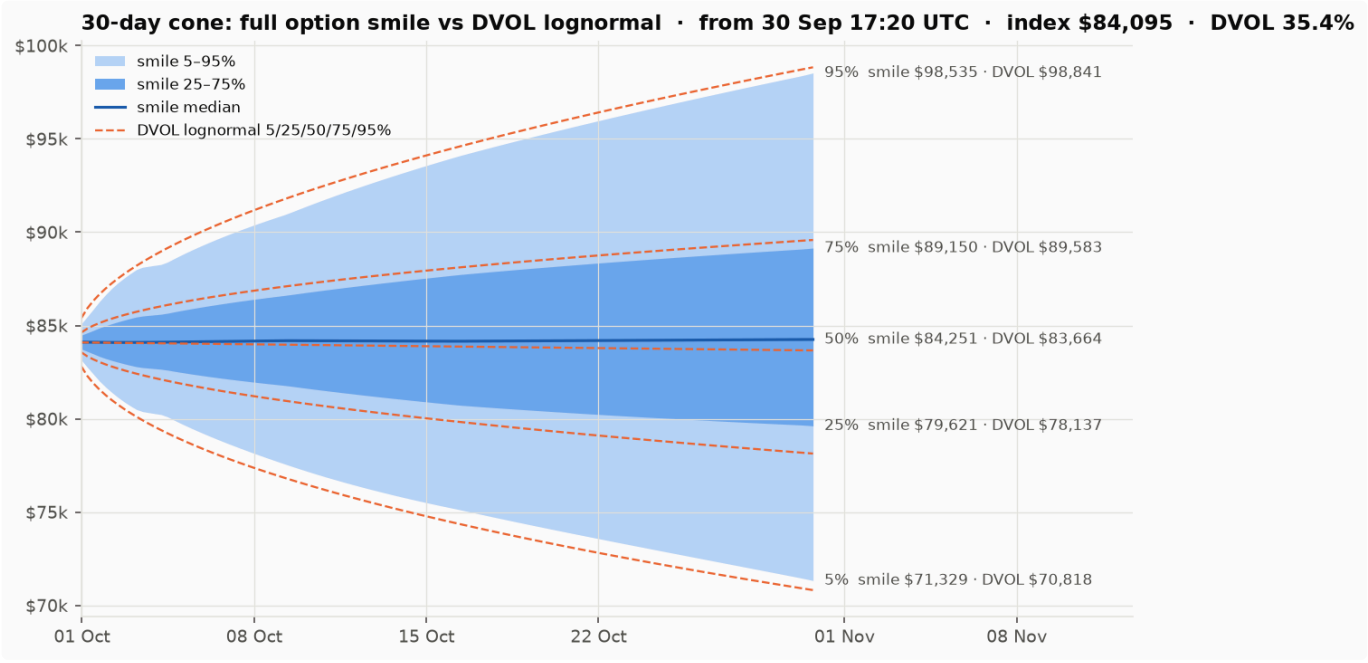
Risk-neutral density by days ahead. From the live snapshot of 30 Sep 2026: smile densities per expiry, interpolated in total variance, with the 5th, 50th and 95th percentiles. (phase3/density_heatmap.png)



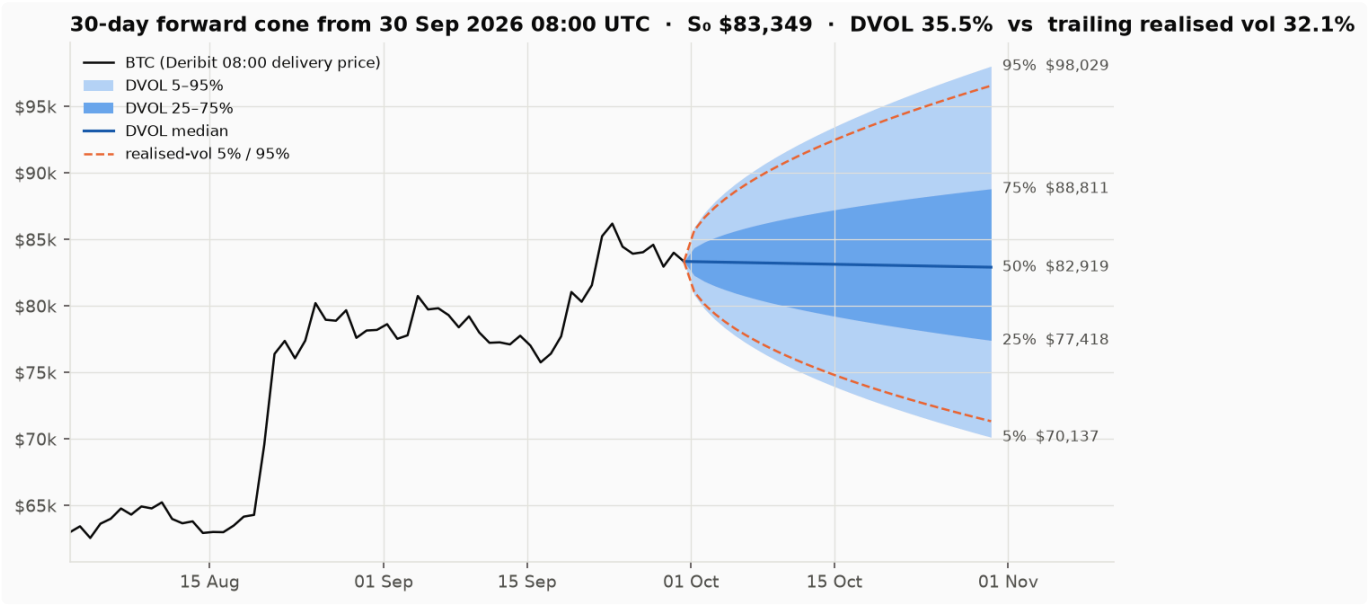
SVI smile fits per expiry. Implied vols of out-of-the-money options in the 30 Sep 2026 snapshot with the fitted SVI curve and fit error per expiry. (phase3/smile_fits.png)



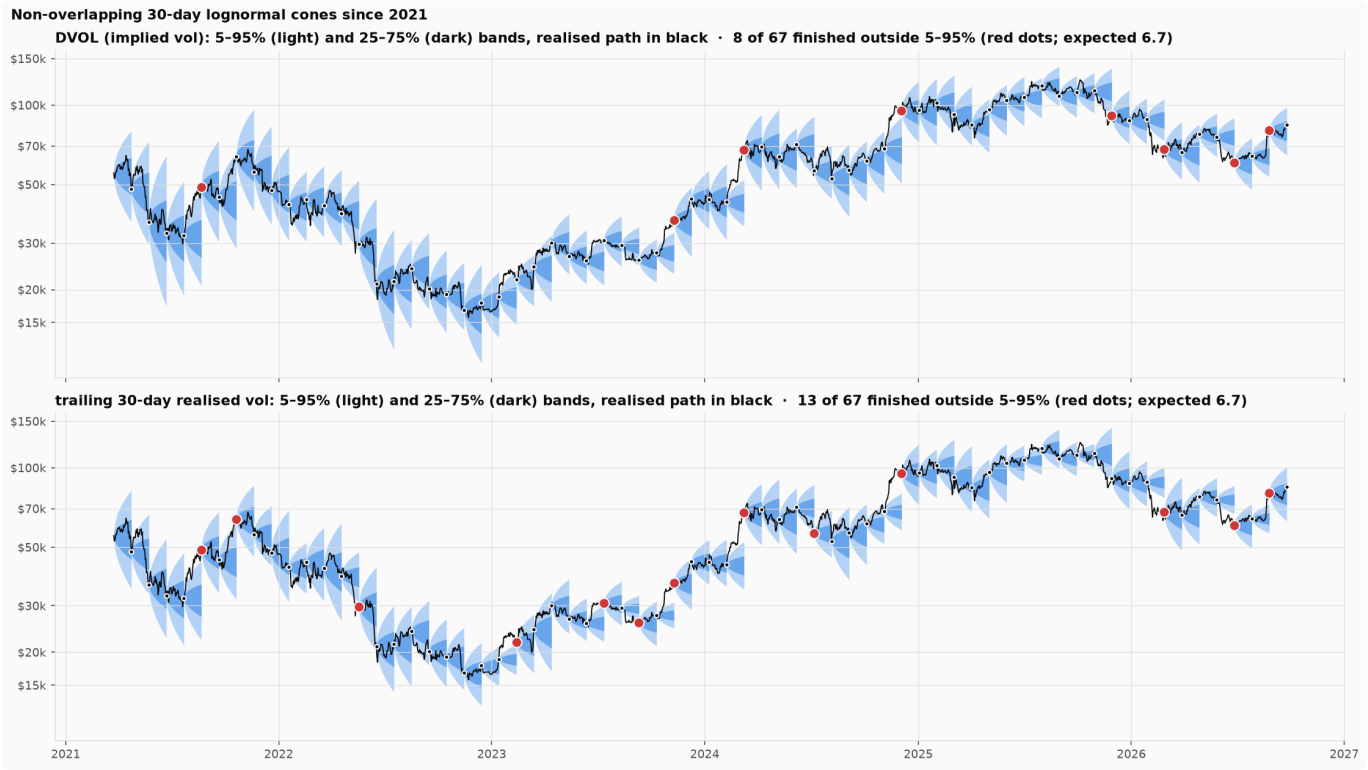
Densities by expiry. Breeden-Litzenberger densities per expiry (30 Sep 2026) against a DVOL lognormal: the smile puts more mass in the centre and the tails. (phase3/expiry_densities.png)



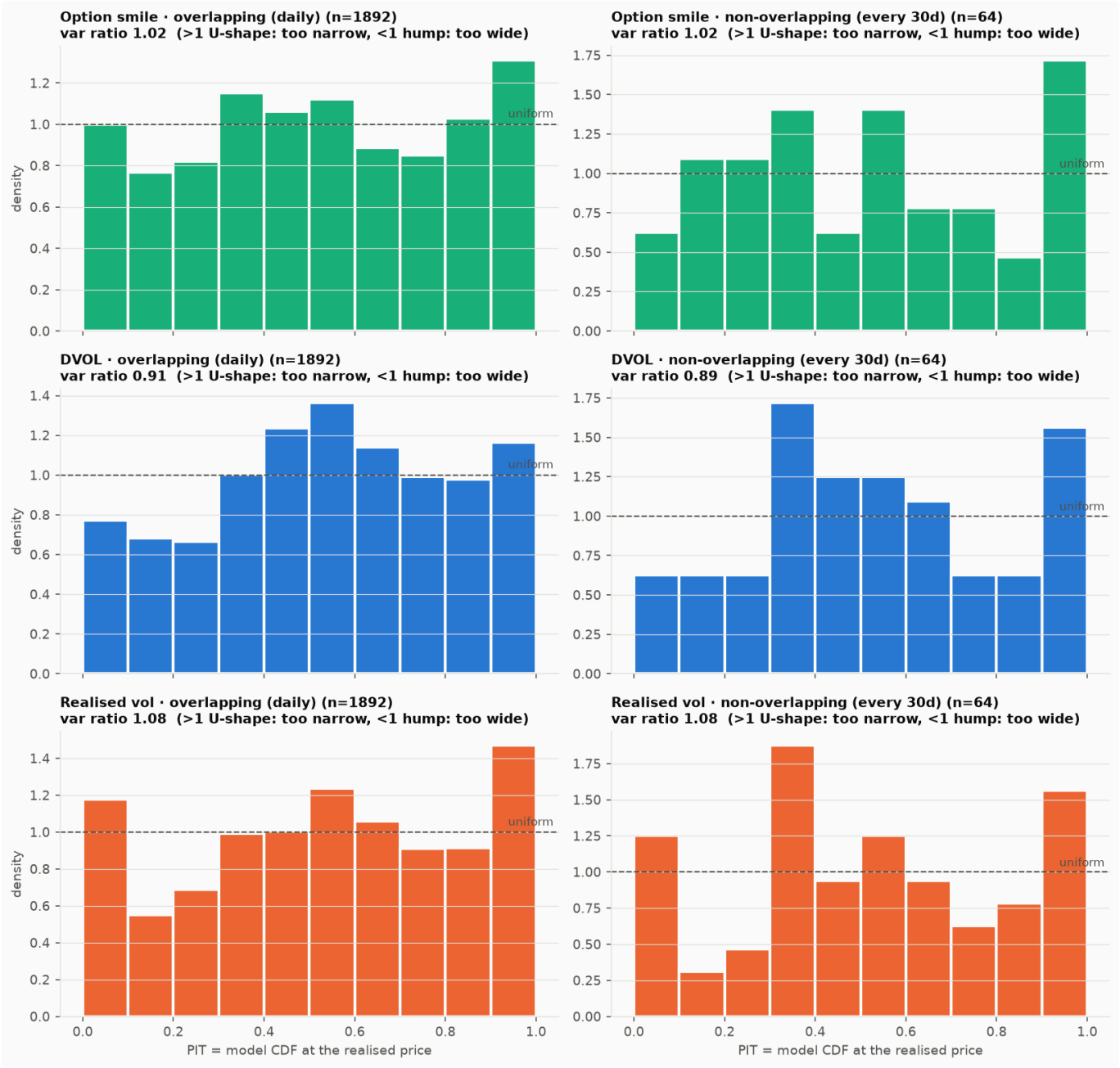
Smile vs DVOL 30-day cone. Percentile bands from the full smile against a DVOL lognormal, from 30 Sep 2026: close at the 5th/95th, the smile is narrower in the middle. (phase3/smile_vs_dvol_cone.png)



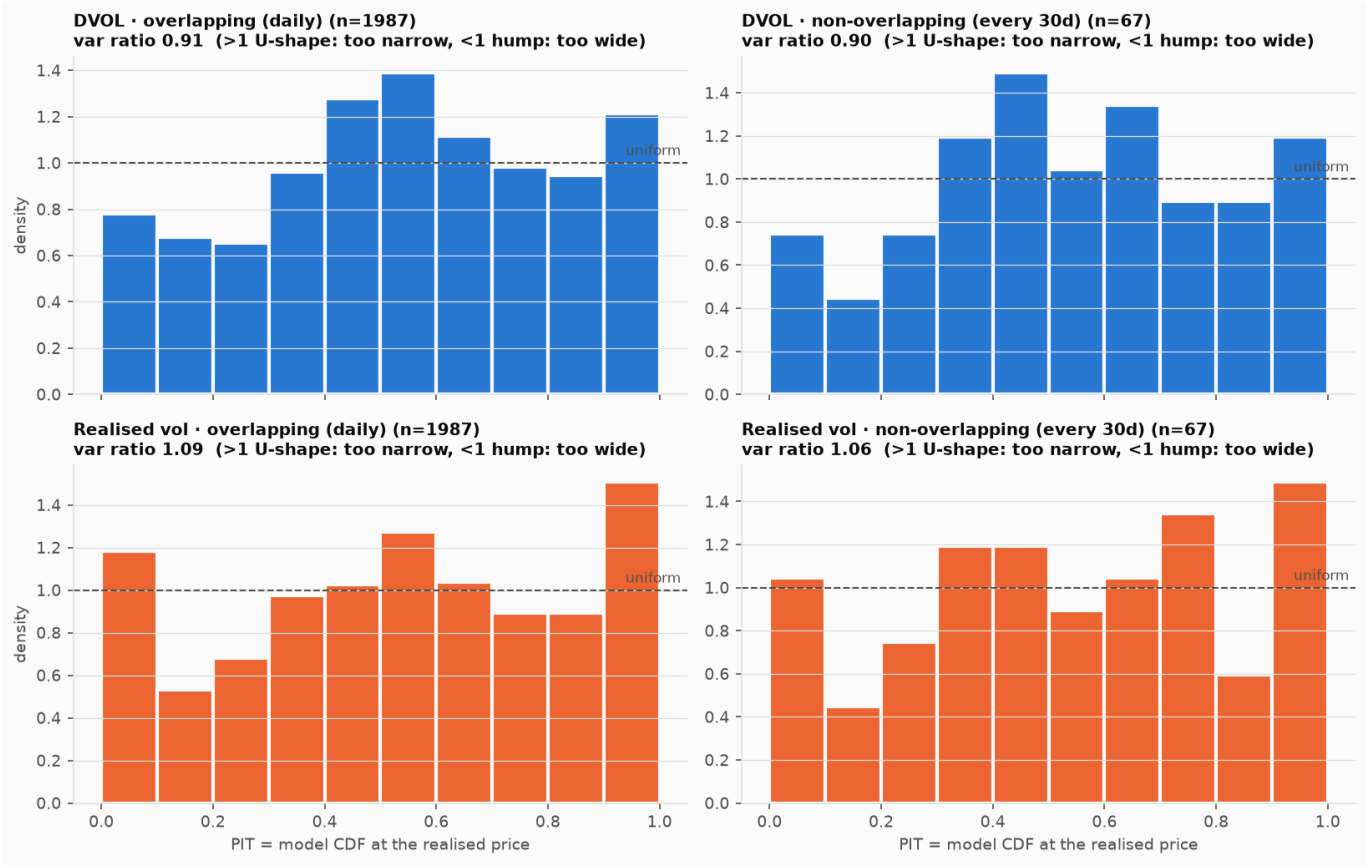
Forward cone. DVOL-based 30-day percentile cone from today. (phase2/forward_cone.png)



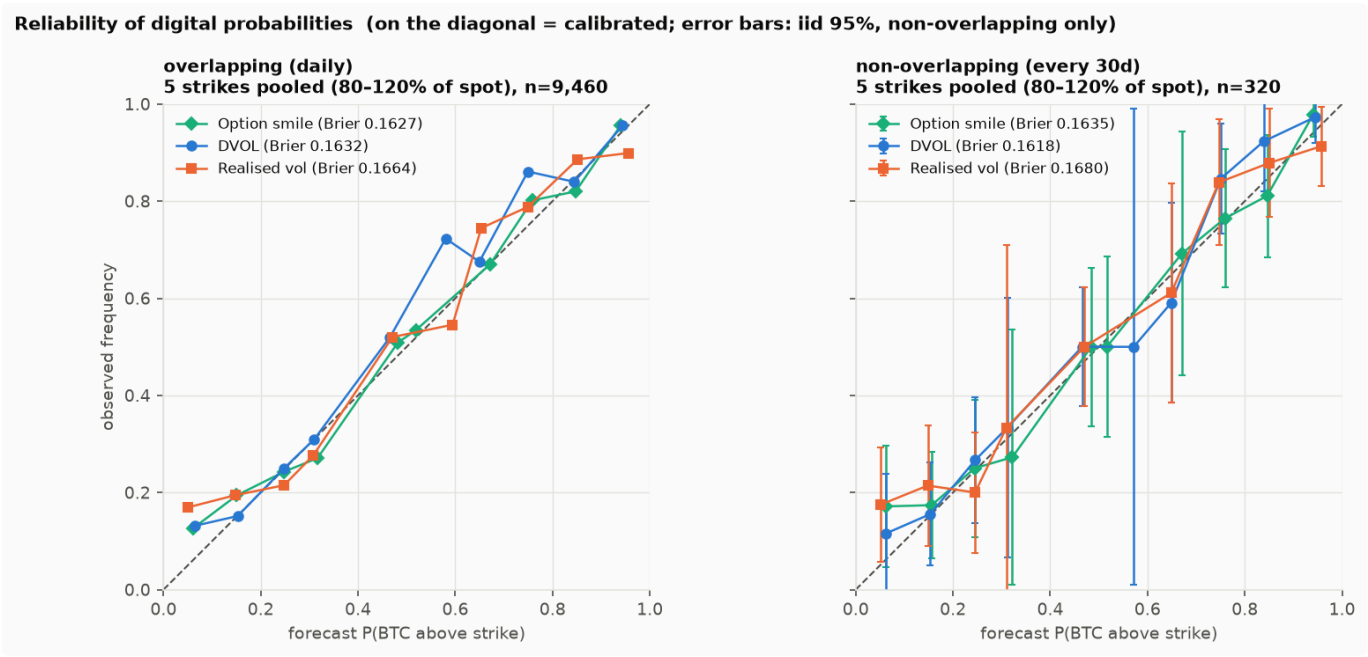
Historical 30-day cones. Non-overlapping 30-day cones since 2021 with the realised path: DVOL 8 of 67 outside 5-95% (expected 6.7), trailing realised vol 13 of 67. (phase2/historical_cones.png)



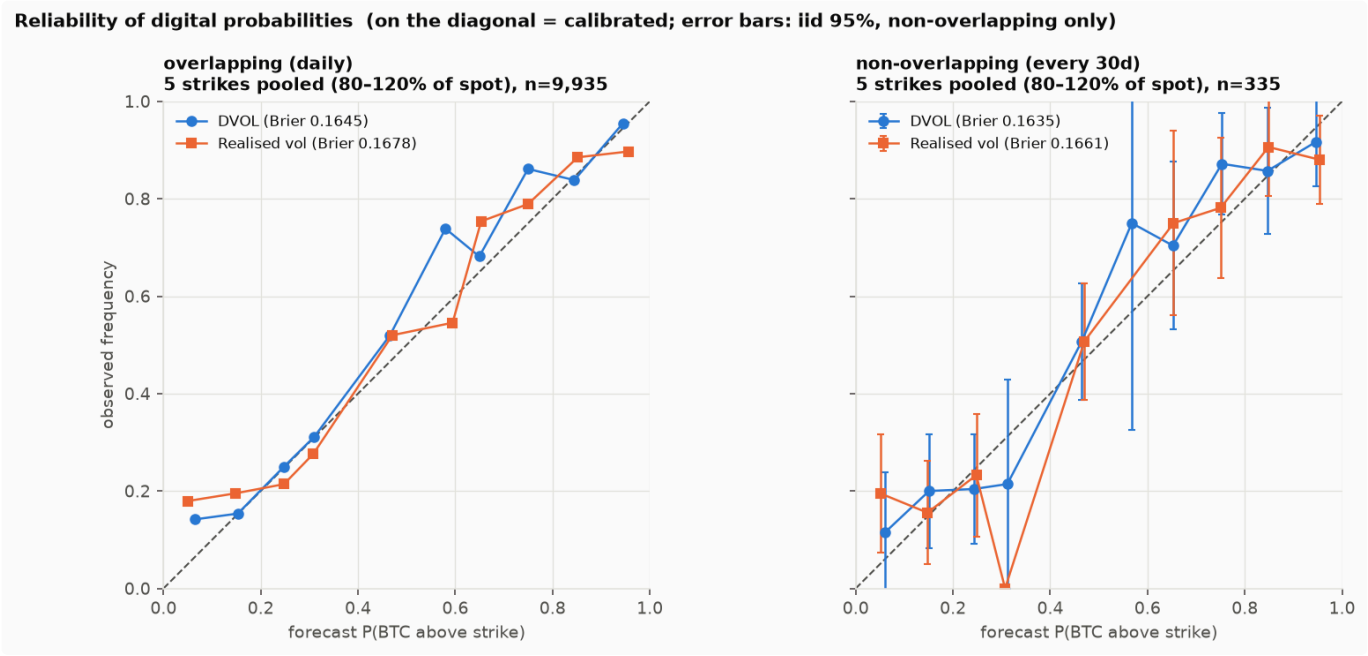
Calibration of the 30-day forecasts (smile). PIT histograms: the smile's are flat (variance ratio 1.02). (phase3/pit_histograms.png)



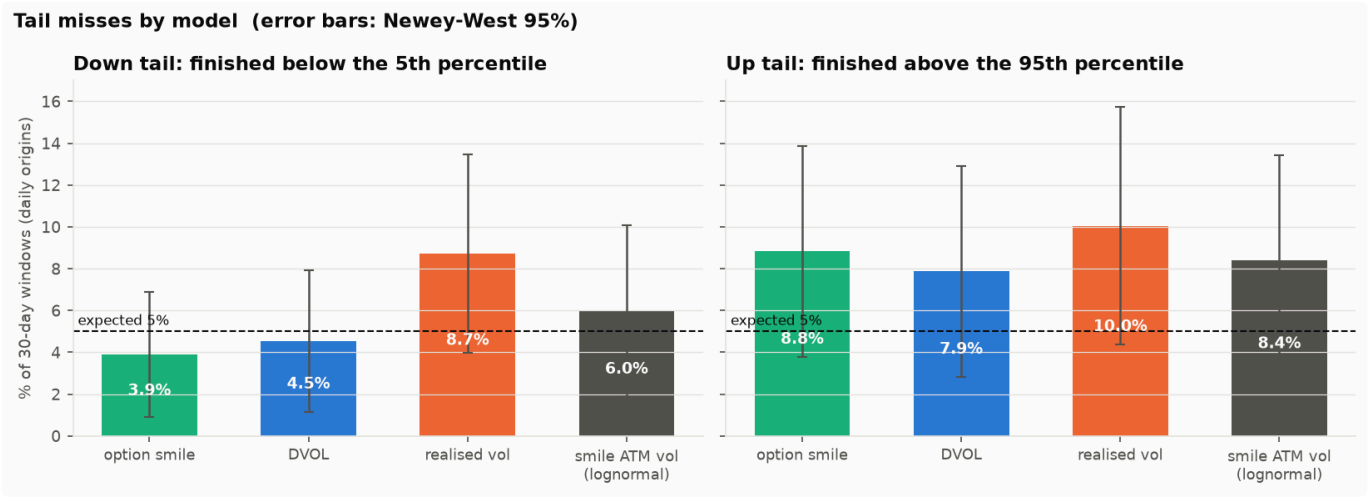
Calibration of the 30-day forecasts (DVOL, realised vol). DVOL is too wide in the middle; realised vol too narrow. (phase2/pit_histograms.png)



Reliability of strike probabilities (smile). Predicted vs observed frequency of BTC ending above each strike. (phase3/reliability.png)



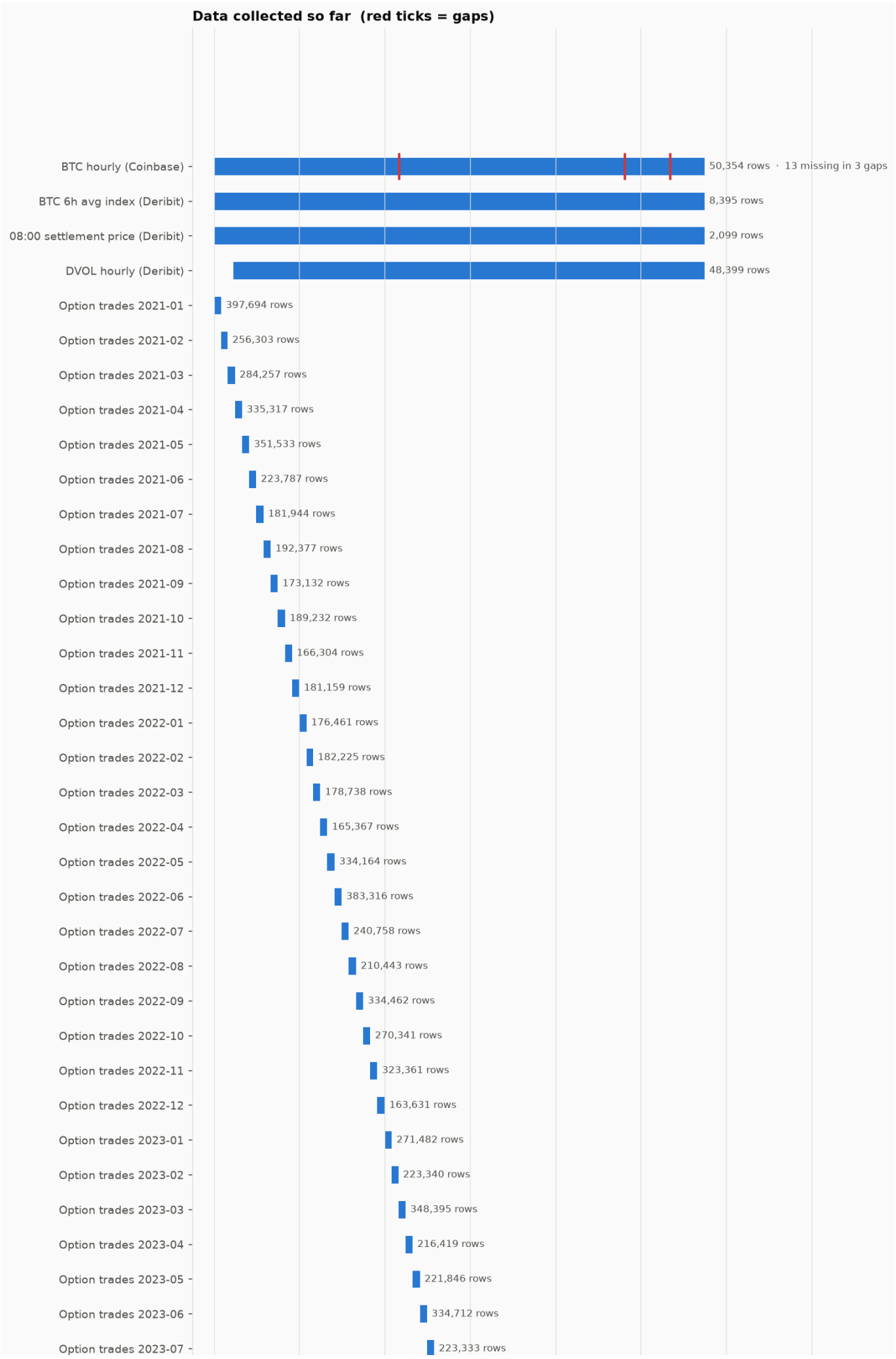
Reliability of strike probabilities (DVOL). The same for the DVOL lognormal and realised-vol models. (phase2/reliability.png)



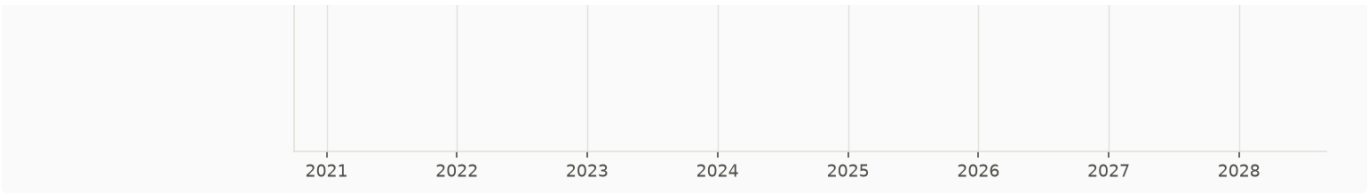
Tail misses by model. Share of 30-day windows finishing below the 5th / above the 95th percentile, by model, with Newey-West 95% CIs: every model misses a little on the upside, none significantly. (phase3/tail_breaches.png)

Data

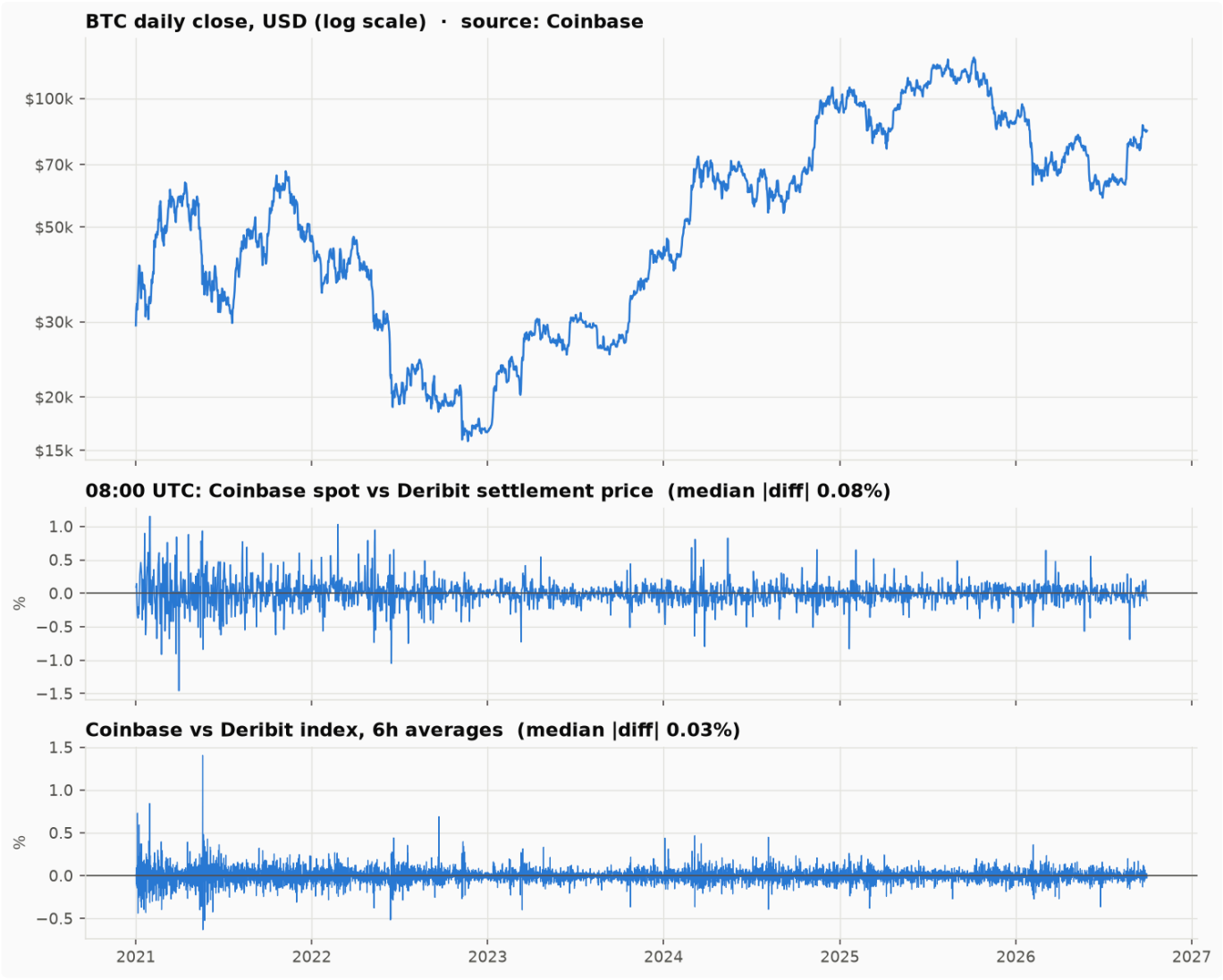
What was collected, and how complete it is.


Data collected so far (red ticks = gaps)
BTC hourly (Coinbase)
50,354 rows · 13 missing in 3 gaps
BTC 6h avg index (Deribit)
8,395 rows
08:00 settlement price (Deribit)
2,099 rows
DVOL hourly (Deribit)
48,399 rows
Option trades 2021-01
397,694 rows
Option trades 2021-02
256,303 rows
Option trades 2021-03
284,257 rows
Option trades 2021-04
335,317 rows
Option trades 2021-05
351,533 rows
Option trades 2021-06
223,787 rows
Option trades 2021-07
181,944 rows
Option trades 2021-08
192,377 rows
Option trades 2021-09
173,132 rows
Option trades 2021-10
189,232 rows
Option trades 2021-11
166,304 rows
Option trades 2021-12
181,159 rows
Option trades 2022-01
176,461 rows
Option trades 2022-02
182,225 rows
Option trades 2022-03
178,738 rows
Option trades 2022-04
165,367 rows
Option trades 2022-05
334,164 rows
Option trades 2022-06
383,316 rows
Option trades 2022-07
240,758 rows
Option trades 2022-08
210,443 rows
Option trades 2022-09
334,462 rows
Option trades 2022-10
270,341 rows
Option trades 2022-11
323,361 rows
Option trades 2022-12
163,631 rows
Option trades 2023-01
271,482 rows
Option trades 2023-02
223,340 rows
Option trades 2023-03
348,395 rows
Option trades 2023-04
216,419 rows
Option trades 2023-05
221,846 rows
Option trades 2023-06
334,712 rows
Option trades 2023-07
223,333 rows

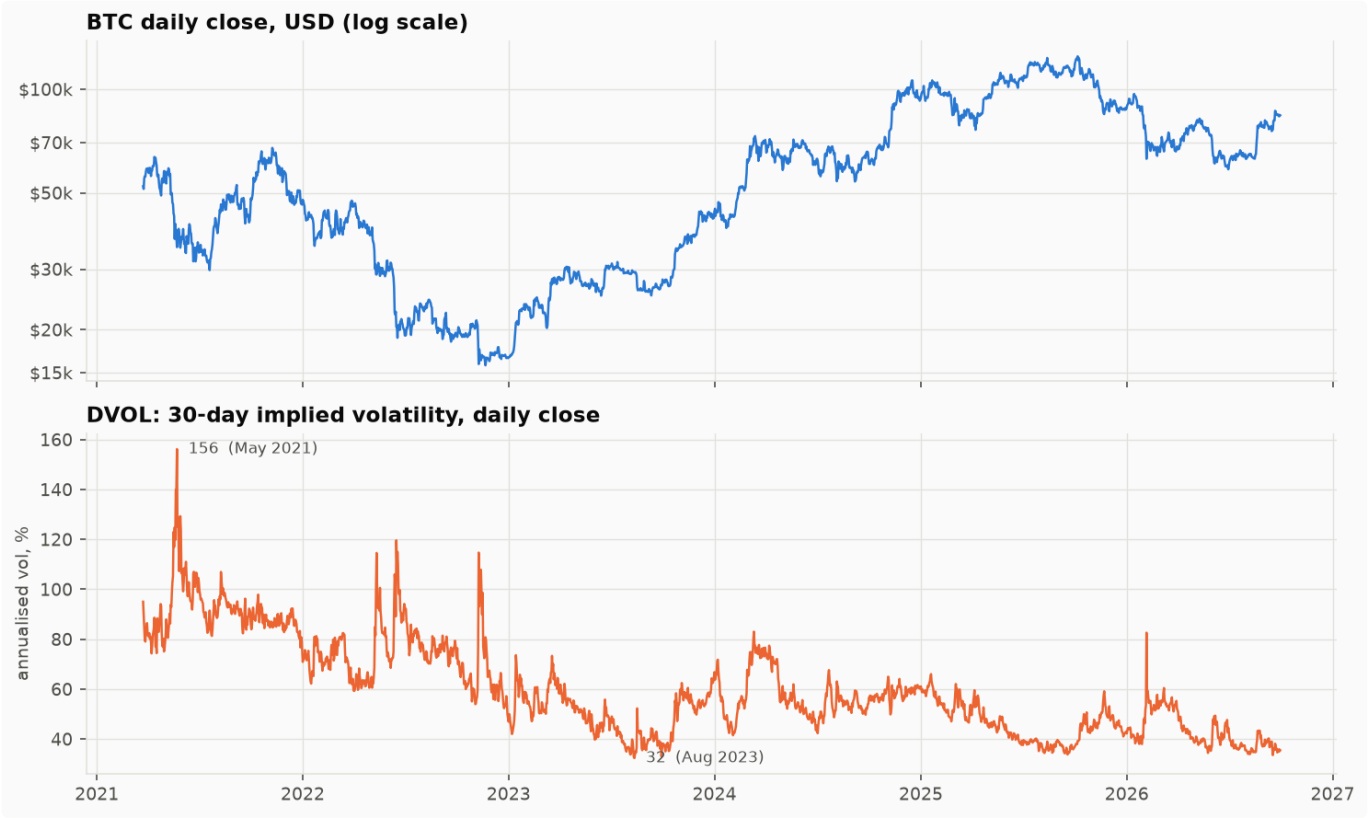

Option trades 2023-08 267,241 rows
Option trades 2023-09 246,526 rows
Option trades 2023-10 383,519 rows
Option trades 2023-11 408,092 rows
Option trades 2023-12 419,078 rows
Option trades 2024-01 436,793 rows
Option trades 2024-02 419,532 rows
Option trades 2024-03 555,316 rows
Option trades 2024-04 416,388 rows
Option trades 2024-05 382,810 rows
Option trades 2024-06 366,848 rows
Option trades 2024-07 444,869 rows
Option trades 2024-08 404,895 rows
Option trades 2024-09 311,727 rows
Option trades 2024-10 345,053 rows
Option trades 2024-11 530,788 rows
Option trades 2024-12 442,554 rows
Option trades 2025-01 447,391 rows
Option trades 2025-02 411,749 rows
Option trades 2025-03 438,055 rows
Option trades 2025-04 492,076 rows
Option trades 2025-05 498,781 rows
Option trades 2025-06 372,753 rows
Option trades 2025-07 427,065 rows
Option trades 2025-08 414,578 rows
Option trades 2025-09 357,247 rows
Option trades 2025-10 578,955 rows
Option trades 2025-11 501,909 rows
Option trades 2025-12 362,695 rows
Option trades 2026-01 358,718 rows
Option trades 2026-02 484,054 rows
Option trades 2026-03 385,432 rows
Option trades 2026-04 299,275 rows
Option trades 2026-05 279,026 rows
Option trades 2026-06 373,832 rows
Option trades 2026-07 280,432 rows
Option trades 2026-08 239,717 rows
Option trades 2026-09 201,712 rows



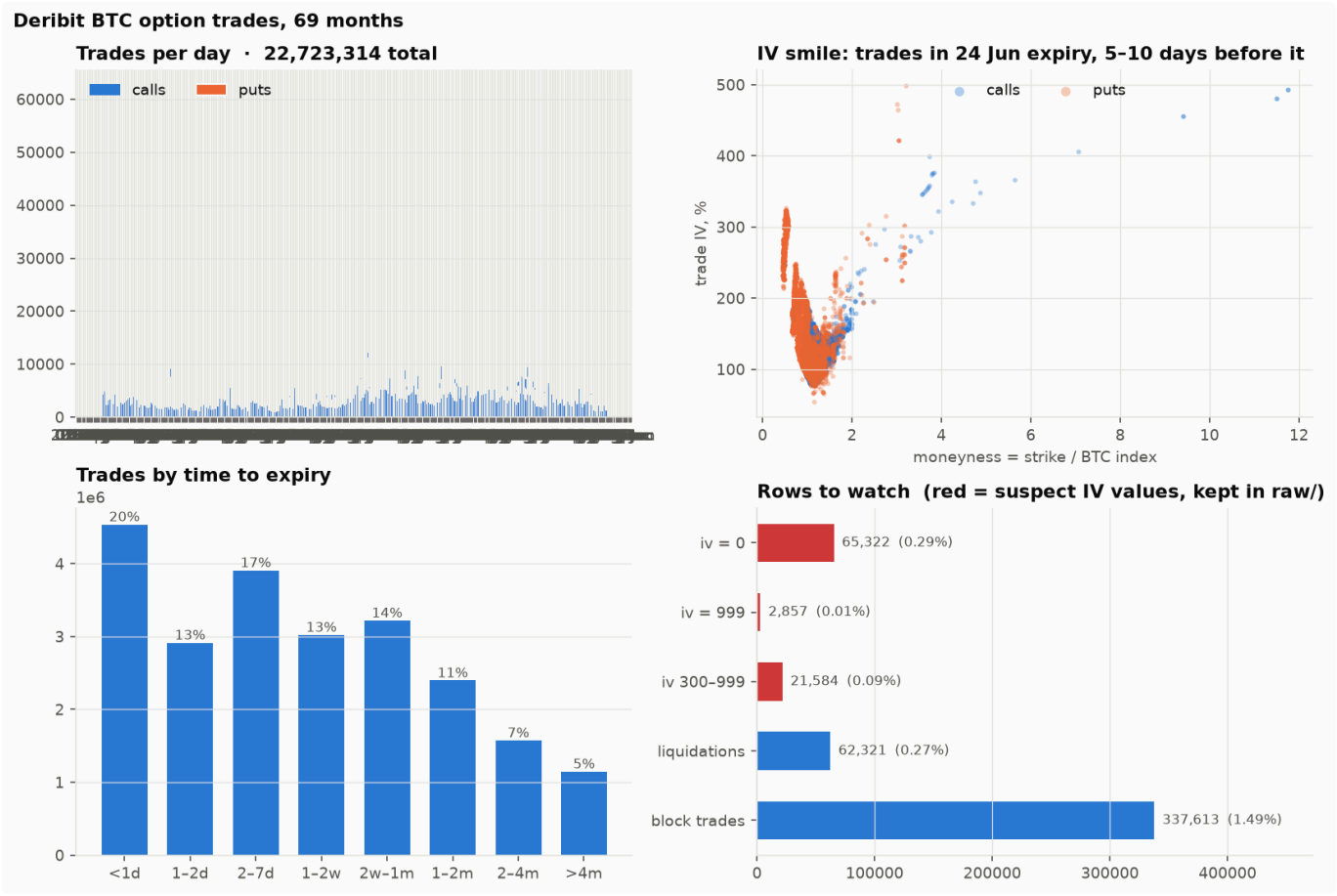
Data coverage. Every dataset's time span and gaps. (figures/00_data_coverage.png)



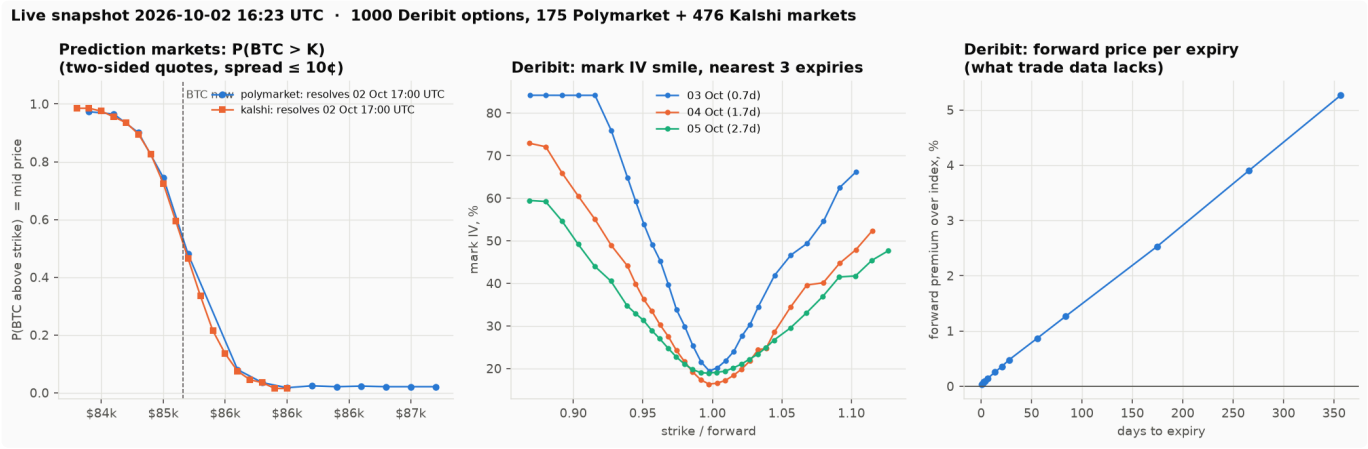
BTC price. Bitcoin price history used throughout. (figures/02_btc_price.png)



BTC vs DVOL. Price against Deribit's 30-day implied volatility index. (figures/03_btc_vs_dvol.png)



Option trades. Deribit option trades per month, 2021-2026. (figures/04_option_trades.png)



A live snapshot. One synchronised snapshot of the Deribit chain and the prediction markets. (figures/05_live_snapshot.png)